Implementing the Marine Strategy Framework Directive (MSFD)

Identifying and addressing social issues and challenges in marine policies in Europe

June 2023

Final report

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The authors of this report are **Clara Jarry**, **Vera Noon** and **Pierre Strosser**, ACT*eon* (France). This report is provided under Contract No 11.0661/2020/828194/SER/ENV.C.2

ACTeon, 5 Place Sainte Catherine, 68000 Colmar – France

E-mail: c.jarry@acteon-environment.eu ; v.noon@acteon-environment.eu ; p.strosser@acteon-environment.eu

Web address: www.acteon-environment.eu

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Abbreviations

CBA	Cost-Benefit Analysis
CBD	Convention on Biological Diversity
CEA	Cost-Effectiveness Analysis
CFP	Common Fisheries Policy
CVM	Contingent Valuation Method
EBA	Ecosystem-Based Approach
EC	European Commission
EFRAG	European Financial Reporting Advisory Group
EIA	Environmental Impact Assessment
EMFAF	European Maritime, Fisheries and Aquaculture Fund
ESA	Economic and Social Analysis
EU	European Union
FAO	Food and Agriculture Organisation
FLAG	Fisheries Local Action Group
FTE	Full Time Employment
GES	Good Ecological Status
IC(Z)M	Integrated Coastal (Zone) Management
ILO	International Labour Organisation
IMP	Integrated Maritime Policy
ITQ	Individual Transferable Quota
LAG	Local Action Group
MCA	Multi-Criteria Analysis
MPA	Marine Protected Area
MSFD	Marine Strategy Framework Directive
MSP	Maritime Spatial Planning
MSPD	Maritime Spatial Planning Directive
RFMO	Regional Fisheries Management Organisation
RSC	Regional Sea Convention
OECD	Organisation for Economic Cooperation and Development
OWF	Offshore Wind Farm
PoM	Programme of Measures
POMESA	Programme of Measures and Socio-Economic Analysis
SDG	Sustainable Development Goal
SFPA	Sustainable Fisheries Partnership Agreement
SIA	Social Impact Assessment
STECF	Scientific, Technical and Economic Committee for Fisheries
TURFs	Territorial User Rights for Fisheries
UN	United Nations
UNGPs	United Nations Guiding Principles
WFD	Water Framework Directive
WG	Working Group
WTP	Willingness To Pay

Summary

This report provides an overview of how social issues and challenges have been, and can be, taken into account in a marine context, responding to the interest of members of the Working Group (WG) Programme Of Measures and Economic & Social Analysis (POMESA) of the Common Implementation Strategy (CIS) of the Marine Strategy Framework Directive (MSFD). First, the report identifies the different elements and concepts that can be considered as being part of social issues and challenges, including social acceptability, equity, inclusion or impacts on socially vulnerable groups. Second, the report presents a state of play on how the social issues and challenges relevant to marine policy are taken into account in different contexts and at different geographical scales, with a particular focus on the implementation of the MSFD. It highlights the primary attention given in the MSFD implementation process to awareness raising, education, information and consultation. Finally, the report proposes a number of operational recommendations to better consider social issues and challenges in the MSFD implementation, building in particular on selected case studies illustrating how social issues have been addressed.

1. Introduction and objectives of the study

Although the Marine Strategy Framework Directive is primarily an environmentally focused policy, it refers to social issues in different parts of the MSFD text. In particular, the MSFD states that Member States (MS) shall[1]:

- Undertake *"an analysis of the features or characteristics of, and pressures and impacts on, their marine waters, identifying the predominant pressures and impacts on those waters, and an economic and **social** analysis of their use and of the cost of degradation of the marine environment"* (Recital 24);
- Conduct *"an economic and **social** analysis of the use of their waters and of the cost of degradation of the marine environment"* (Article 8);
- *"Give due consideration to sustainable development and, in particular, to the **social** and economic impacts of the measures envisaged"* when drawing up their programme of measures (PoM) (Article 13);
- Take *"due consideration of **social** and economic concerns in the setting of targets"* (Annex IV).

Social aspects should therefore be considered at different stages of the MSFD implementation process as they are being legally referred to. Taking them into account in practice requires however that what lays behind the term "social" is better defined and made explicit.

Social issues have received limited attention to date in the MSFD implementation process at different (national, regional and EU) scales. However, there is increasing interest in social issues, as illustrated by a **survey** sent to members of the Programme of Measures and Economic & Social Analysis working group (POMESA WG) that aimed at assessing current practices in the application of socio-economic assessments to support the MSFD implementation and areas of priority interest for collective work[2]. Answers of POMESA WG members to the survey indicated interest in further investigating social issues, in particular in relation to:

- *The impact of measures at the socio-economic level*;
- *Social indicators and ecosystem services for public policy*;
- *Public awareness raising (so the complexity of the marine environment is better understood) and research*;
- *Employment*;
- *Social issues in general, e.g. integrating them in cost benefit analyses*;
- *Blue jobs, blue growth, minority/equality issues, fairness*;
- *Distribution of benefits and costs*;
- *Public understanding of eutrophication and the role of agriculture, as well as social impacts/challenges related to the shrinking of fisheries*;
- *People's perception on challenges around the marine environment*.

Building on this shared interest, the present study aims at shedding light on the different social aspects that can be considered in relation to the implementation of the MSFD and other relevant marine policies, strategies and initiatives. Carried out in the context of a service contract that supports the European Commission in the implementation of the MSFD, it focuses in particular on:

[1] Marine Strategy Framework Directive, 2008. EUR-Lex - 32008L0056 - EN - EUR-Lex (europa.eu)

[2] ACTeon, 2021. Assessing current practice with the application of socio-economic assessments to support marine policy/MSFD implementation.

- **Identifying the different social issues and challenges** faced in different regional seas and at different decision-making scales;
- Analysing how these social issues and challenges are **considered in the practical implementation** of marine policies, strategies, measures and projects – in terms of assessments carried out, decisions/prioritisation of actions, the design of stakeholder processes, financing, etc.;
- Identifying **best practices and recommendations** for better internalising social issues and challenges in the MSFD implementation (in particular in designing stakeholder processes and carrying out ESA), building on an overview of how various countries have been dealing with these issues.

2. Identifying different social issues and challenges

Working on social issues and challenges raises a major question: how can "social aspects" be defined, in particular in the marine sector, and then, how can these be assessed? Indeed, the term "social" is subject to many different interpretations depending on the disciplines and policies considered. There are several official and guidance documents, notably established at the UN and EU levels, dealing with social aspects in a general manner (without necessarily focusing on the ocean). These documents include (not exhaustive):

- The UN Declaration on the Rights of Indigenous Peoples[3];
- The UN SDGs[4] related to socio-economic issues with their associated targets and indicators;
- The European Convention on Human Rights[5];
- The European Social Charter[6];
- The EU Charter on Fundamental Rights[7];
- The document on the European pillar of social rights[8] and its associated action plan, providing 20 social principles.

These documents provide the basis for the identification of social issues that can be relevant to marine policy implementation and ocean management, social aspects considered in each document being presented in Annex 1 of this report. Several documents also address employment issues (in particular labour rights and working conditions in the fisheries sector), including from:

- The International Labour Organisation (ILO) Maritime Labour Convention[9];
- The ILO Work in Fishing Convention[10];
- The Food and Agriculture Organisation (FAO) Code of Conduct for Responsible Fisheries[11];
- The FAO Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries[12];
- The FAO Guidance framework on social responsibility in fisheries and aquaculture value chains[13];
- The FAO Voluntary Guidelines on the Responsible Governance of Tenure[14].

Other initiatives such as **ocean accounts** contribute to the coherent measurement of progress towards the sustainable development of the ocean, including from a social perspective. Ocean accounts consist

[3] United Nations Declaration on the Rights of Indigenous Peoples | United Nations For Indigenous Peoples (Last viewed on 29/08/2022)
[4] https://sdgs.un.org/ (Last viewed on 29/08/2022)
[5] European Court of Human Rights, 2021. European Convention on Human Rights. European Convention on Human Rights (coe.int)
[6] The European Social Charter (coe.int) (Last viewed on 11/08/2022)
[7] EU Charter of Fundamental Rights | European Commission (europa.eu) (Last viewed on 11/08/2022)
[8] The European Pillar of Social Rights in 20 principles | European Commission (europa.eu) (Last viewed on 11/08/2022)
[9] International Labour Organisation, 2006. Maritime Labour Convention. wcms_763684.pdf (ilo.org)
[10] International Labour Organisation, 2007. Work in Fishing Convention. Convention C188 - Work in Fishing Convention, 2007 (No. 188) (ilo.org)
[11] FAO, 1995. Code of Conduct for Responsible Fisheries. FAO Code of Conduct for Responsible Fisheries
[12] SSF Guidelines | Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries | Food and Agriculture Organization of the United Nations (fao.org) (Last viewed on 09/09/2022)
[13] FAO, 2019. Social responsibility in fish value chains. nb389en.pdf (fao.org)
[14] Voluntary Guidelines on Tenure | Governance of Tenure | Food and Agriculture Organization of the United Nations (fao.org) (Last viewed on 09/09/2022)

in integrated records of data on ocean environmental attributes, economic activity and social considerations[15].

In the context of this study, the main social issues and challenges related to marine policies and strategies have been categorised into three main blocks[16]:

- **Social impacts**;
- **Measures and actions that help capturing or dealing with social challenges**;
- **Governance and decision-making processes**.

The study addresses most of the topics of interest to POMESA members, with less attention paid to the issue of blue growth (a topic with a strong economic component) and the role of agriculture in eutrophication (despite the undeniable connection between agriculture, eutrophication and marine Good Ecological Status, European marine policies having still limited scope for action on agriculture-related issues).

The key principles of **fairness**, **inclusiveness** and **social acceptability/acceptance** are considered overarching issues considered transversally for each of the three blocks[17].

2.1. Defining key principles of fairness, inclusiveness and social acceptability/acceptance

2.1.1. Fairness and inclusiveness

The term "**fairness**" is often connected with the terms of "justice" and "equity". However, their use differs between disciplines. In social sciences and conservation, the three terms are usually used interchangeably. When it comes to environmental governance focusing on stakeholders' perceptions, "fairness" and "justice principles" are considered differently. "Fairness" is used *"to describe individuals' perceptions arising from judgements of situations related to dimensions of justice (e.g. distribution, procedure)"*, whereas "justice principles" (e.g. equality) are considered to exist *"independently of any process of judgement"*[18]. Fairness can be approached from several angles, such as the distribution of access to and benefits from ecosystem services ("distributional justice") or the engagement in decision-making processes. Equality can also be seen as a form of justice or fairness, but these cannot be limited to this sole principle. To go further, three major distributional justice principles of "need", "equality" and "proportionality" have been identified and described as follows: *"the 'need' principle focuses on the scale of individuals' needs and suggests that individuals who are most dependent on the benefits under consideration should receive the largest share, or alternatively, those who have difficulty in dealing with costs, should bear the least*[19]*. The 'equality' principle relates to a uniform distribution of cost and benefits based on the notion that everybody should be treated alike. Lastly, the*

[15] What are ocean accounts? (Last viewed on 13/02/2023)

[16] This categorisation into these three blocks is somewhat arbitrary, with some overlaps and synergies between the blocks.

[17] Are all stakeholders (including vulnerable groups, young generation, women, local and indigenous communities…) and sectors equally impacted, supported and involved in decision-making processes, thus contributing to social acceptability/acceptance?

[18] GURNEY G. G., MANGUBHAI S., FOX M., KIATKOSKI KIM M. and AGRAWAL A., 2021. Equity in environmental governance: perceived fairness of distributional justice principles in marine co-management. Environmental Science & Policy, 124, 23-32. doi:10.1016/j.envsci.2021.05.022

[19] These terms are presented as defined in the literature. However, their practical application remains a political decision, including in how they can be balanced with other principles such as the Polluter Pays Principle and whether compensation to costs should take place or not.

'proportionality' principle focuses on the balance between input and output, whereby the benefits an individual receives should be proportional to the costs they bear" or the effort (in addressing a given problem that can be socially expected and/or justified[20] The fairness of how different benefits are distributed, or of the cost-sharing mechanisms put in place, can be perceived differently by different social groups, depending on individuals' experiences and their socio-cultural context.

The UN's Department of Economic and Social Affairs defines **social inclusion** (which can also be referred to as social inclusiveness or integration) as *"the process by which efforts are made to ensure equal opportunities – that everyone, regardless of their background, can achieve their full potential in life. Such efforts include policies and actions that promote equal access to (public) services as well as enable citizen's participation in the decision-making processes that affect their lives"*[21].

Fairness and inclusiveness are intertwined principles that are often addressed concurrently. Several studies show these two principles are not given sufficient attention in the pursuit of the sustainable management and protection of the ocean. The difficulties reported in relation to the **establishment of Marine Protected Areas (MPAs)**, for instance, is often associated with a lack of inclusion, failure to consider local people's needs and livelihoods and the dispossession of areas and resources. Fisheries management decisions are sometimes criticised for their lack of consideration of equity and of distributional impacts, with the needs and livelihoods of small-scale fishers and coastal communities not being taken into account. Similar concerns have been raised regarding the impacts of the development of the **blue economy** in general: blue economic developments may fail to benefit local people, and when they do so, benefits are often shared inequitably between local communities and interests behind developments. Social and environmental burdens (e.g. waste, water shortages, pollution…) may also considerably affect nearby communities. The inequitable distribution of benefits and costs may be explained, inter alia, by the lack of consultation or engagement in decisions. Fishers and coastal communities are among the first to be impacted by climate change, but climate change adaptation processes often fail to include local people in decision-making, therefore producing adaptations that can further marginalise already vulnerable groups[22].

When it comes to marine conservation sciences, there are many barriers to justice, equity, diversity and inclusion limiting the success of marine conservation professionals from under-represented and marginalised groups (e.g. women, LGBTQ+, Indigenous and people of colour). The main challenges identified in marine sciences include lack of leadership roles for underrepresented groups, access to academic conferences and representation across all communities[23].

2.1.2. Social acceptability and acceptance

Social acceptance can be defined in relation to the public response to a project or measure, which can be interpreted as social acceptance if positive, whereas it can be referred to as public opposition if

[20] GURNEY G. G., MANGUBHAI S., FOX M., KIATKOSKI KIM M. and AGRAWAL A., 2021. Equity in environmental governance: perceived fairness of distributional justice principles in marine co-management. Environmental Science & Policy, 124, 23-32. doi:10.1016/j.envsci.2021.05.022

[21] Social Inclusion | Poverty Eradication (un.org) (Last viewed on 25/08/2022)

[22] BENNETT, N.J., 2018. Navigating a just and inclusive path towards sustainable oceans. Marine Policy. https://doi.org/10.1016/j.marpol.2018.06.001

[23] JOHRI S., CARNEVALE M., PORTER L., ZIVIAN A., KOURANTIDOU M., MEYER E. L., SEEVERS J. and SKUBEL R. A., 2021. Pathways to Justice, Equity, Diversity, and Inclusion in Marine Science and Conservation. Frontiers in Marine Science. Volume 8, Article 696180. doi: 10.3389/fmars.2021.696180

negative. Social acceptance suggests a degree of tolerance rather than necessarily willingness and contentment of approval[24]. Social acceptance is critical to the development of projects in marine areas. Public opposition is one of the main reasons for the delay in project development, notably in the case of offshore wind projects[25]. This is especially true for the position of economic sectors (e.g. fisheries) that may be impacted by these projects. On the other hand, opposition from citizens may be less than in the case of land-based projects. Social acceptance is also crucial for MPAs, where success highly depends on the management regime and the enforcement of the restrictions that apply in the MPA[26]. Those are likely to be a reflection of social, cultural, economic and political factors, which seem to have a more important role than the biological or physical considerations that support the MPA success[27]. In the context of MPAs, Thomassin *et al.* (2010)[28] have defined social **acceptability** as *"a measure of support towards a set of regulations, management tools or towards an organisation by an individual or a group of individuals based on geographic, social, economic or cultural criteria"*.

In the context of renewable energy, Wüstenhagen *et al.* (2007)[29] have distinguished social acceptance into three dimensions: "**socio-political** acceptance", which refers to the acceptance by major social actors and general public; "**community** acceptance", which refers to the acceptance by local stakeholders and communities; and "**market** acceptance", which translates the willingness of consumers, investors and businesses to "enter" and support the market of a given product or service. There may be discrepancies between the general public and the community opinions since the general public is in the majority of cases not strongly connected to affected areas[30]. In particular, it is at the community level that issues related to NIMBYism (*Not in my backyard*) are expected to arise[31]. Social acceptance at a public level may therefore coexist with unacceptance at community level[32].

Communication between local communities, decision-makers and developers in the case of a project as well as cooperation among different actors are key to social acceptance. Public engagement throughout the project development or decision process, i.e. when affected people are involved prior

[24] GARZON, A., RABUTEAU, Y. and STANLEY, K., 2013. Civil society involvement and social acceptability of marine energy projects – Best practices of the marine energy sector. A report prepared as part of the MERIFIC project "Marine Energy in Far Peripheral and Island Communities". https://tethys.pnnl.gov/sites/default/files/publications/MERiFIC-6.1.2-Civil-society-involvement.pdf

[25] SOMA, K., HAGGETT, C., 2015. Enhancing social acceptance in marine governance in Europe. Ocean & Coastal Management, 117, 61–69. doi:10.1016/j.ocecoaman.2015.11.001

[26] SPAANS, M., 2020. Marine Protected Areas in Europe – An assessment of the current status and the representation of benefits in socio-economic analyses in order to support decision-making. https://www.noordzeeloket.nl/publish/pages/184490/status_and_benefits_of_marine_protected_areas_in_europe_final.pdf

[27] GALL, S. C., RODWELL, L. D., 2016. Evaluating the social acceptability of Marine Protected Areas. Marine Policy, 65(), 30–38. doi:10.1016/j.marpol.2015.12.004

[28] THOMASSIN A., WHITE C. S., STEAD S. S., DAVID G., 2010. Social acceptability of a marine protected area: The case of Reunion Island. , 53(4), 169–179. doi:10.1016/j.ocecoaman.2010.01.008

[29] WÜSTENHAGEN, R., WOLSINK, M., BÜRER, M.J., 2007. Social acceptance of renewable energy innovation: an introduction to the concept. Energy Policy 35, 2683-2691. http://dx.doi.org/10.1016/j.enpol.2006.12.001.

[30] SOMA, K., HAGGETT, C., 2015. Enhancing social acceptance in marine governance in Europe. Ocean & Coastal Management, 117, 61–69. doi:10.1016/j.ocecoaman.2015.11.001

[31] GARZON DELVAUX, P. A., RABUTEAU, Y. and STANLEY K., 2013. Civil society involvement and social acceptability of marine energy projects – Best practices of the marine energy sector. Report prepared as part of the MERIFIC project ("Marine Energy in Far Peripheral and Island Communities".

[32] WÜSTENHAGEN, R., WOLSINK, M., BÜRER, M.J., 2007. Social acceptance of renewable energy innovation: an introduction to the concept. Energy Policy 35, 2683-2691. http://dx.doi.org/10.1016/j.enpol.2006.12.001.

to a policy decision, can shape social acceptance. It can then be considered that social acceptance at community level depends on three core dimensions[33]:

- Actual intervention – What is it and what is acceptable?
- Public engagement – How is development generated through processes of engagement?
- Authority – Who is entitled to make rules?

There is often no clear distinction made between social "acceptance" and "acceptability", some authors distinguishing the two while others use both terms interchangeably. Some authors like Fournis and Fortin (2016)[34] distinguish social acceptance and social acceptability. Social acceptance is *"seen as one of the possible results (vs. unacceptance) of a complex process of social acceptability"*, the latter being a broader concept. They define social acceptability as *"the process of collective assessment of a given project, integrating a plurality of actors (stakeholders) and spatial scales (from global to local), as well as involving the specific trajectory (past, present and future) of a political group or polity (community/society)"*.

The notions of social acceptance and acceptability are closely linked with fairness and inclusiveness as well. Lack of inclusion in decision-making and perceived or real inequities can lead to local community opposition to projects or measures such as the establishment of MPAs or the development of blue economy projects, therefore undermining the ocean sustainability agenda[35].

2.2. Categorising social issues into three blocks

Social issues and challenges can be categorised into three main blocks: the social impacts of marine policies; (political) measures and actions that help capturing or dealing with social issues; and governance and decision-making processes. These blocks encompass most of the topics of interest to POMESA members, with some of the topics being more scrutinised in case studies.

2.2.1. Analysis of social impacts

The latest amendment of the **EU Environmental Impact Assessment (EIA) Directive** (2014/52/EU)[36] focuses in priority on biophysical and ecological impacts, the socio-economic dimension being limited to population, human health and cultural heritage[37].

[33] SOMA, K., HAGGETT, C., 2015. Enhancing social acceptance in marine governance in Europe. Ocean & Coastal Management, 117, 61–69. doi:10.1016/j.ocecoaman.2015.11.001

[34] FOURNIS, Y. and FORTIN, M.-J., 2016. From social 'acceptance' to social 'acceptability' of wind energy projects: towards a territorial perspective, Journal of Environmental Planning and Management, doi:10.1080/09640568.2015.1133406

[35] BENNETT, N.J., 2018. Navigating a just and inclusive path towards sustainable oceans. Marine Policy. https://doi.org/10.1016/j.marpol.2018.06.001

[36] Directive 2014/52/EU of the European Parliament and of the Council of 16 April 2014 amending Directive 2011/92/EU on the assessment of the effects of certain public and private projects on the environment. EUR-Lex - 32014L0052 - EN - EUR-Lex (europa.eu)

[37] GLASSON, J., DURNING, B., OLORUNDAMI, T. and WELCH, K., 2020. Guidance on assessing the socio-economic impacts of offshore wind farms (OWFs). Guidance on assessing the socio-economic impacts of offshore wind farms (OWFs) | openEQUELLA (brookes.ac.uk)

The **European Commission (EC) Better Regulation Toolbox**[38], a guidance document related to impact assessment, gives insight in the potential social impacts that might be induced by (new) legislation. It sets out categories of social impacts to be considered when conducting an impact assessment. The tool #30 covers three categories of social impacts that policy options may induce: *"impacts on 1) employment, 2) working conditions and 3) income distribution, social protection and inclusion, including impacts on perspectives and life-situations of people in, or at risk of poverty"*. Particular attention should be given to stakeholders who often take less part in discussions, such as vulnerable groups with disadvantaged backgrounds (e.g. people with migration background) or facing multiple disadvantages (e.g. women living in rural areas), in accordance with the fairness and inclusiveness principles. Other tools, in particular tools #29, #31 and #32 respectively on "fundamental rights, including the promotion of equality", "education and training, culture and youth (ETCY)" and "health impacts", are also of relevance when analysing social issues l.

The **International Association for Impact Assessment** has also produced a guidance document for assessing the social impacts of projects[39]. Social Impact Assessment (SIA) can be described as *"the process of identifying and managing the social issues of project development, and includes the effective engagement of affected communities in participatory processes of identification, assessment and management of social impacts"*. Social impacts can be changes to one or more of the following[40]: people's way of life; their culture; their community; their political systems; their environment; their health and well-being; their personal and property rights; or their fears and aspirations. Social impacts can be both positive and negative, with SIA aiming to identify solutions or accompanying measures that can enhance the (social) benefits of projects to impacted communities.

In the case of the implementation of a marine project, e.g. Offshore Wind Farms (OWFs), social impacts include impacts on the demography and local services, as well as impacts on the socio-cultural and quality of life of the host coastal area. Marine projects may also raise distributional issues, i.e. when they have particularly strong impacts on some societal groups (e.g. local and non-local groups, socio-economic groups). There are some methodological challenges in assessing these social impacts, in particular how to capture, qualify and/or quantify social and cultural values as well as the potential impact of a project on these values[41].

Only social impacts considered the most relevant to assess when focusing on marine-related policies and strategies have been retained for this study. These social impacts can be classified based on the group of stakeholders they affect. In the European Financial Reporting Advisory Group's (EFRAG) final report for developing EU sustainability-reporting standards[42], five categories of stakeholders have been identified, namely workforce, value-chain workers, affected communities, end-users/consumers and wider society. These same groups have also been taken up in the Final Report on Social

[38] European Commission, 2021. Better Regulation Toolbox. Better regulation: guidelines and toolbox | European Commission (europa.eu) (Last viewed on 08/08/2022)

[39] VANCLAY, F., ESTEVES, A.M., AUCAMP, I. and FRANKS, D., 2015. Social Impact Assessment: Guidance for assessing and managing the social impacts of projects. Fargo ND: International Association for Impact Assessment

[40] VANCLAY, F., 2003. International Principles for Social Impact Assessment. Impact Assessment & Project Appraisal 21(1), 5-11. http://dx.doi.org/10.3152/147154603781766491

[41] GLASSON, J., DURNING, B., OLORUNDAMI, T. and WELCH, K., 2020. Guidance on assessing the socio-economic impacts of offshore wind farms (OWFs). Guidance on assessing the socio-economic impacts of offshore wind farms (OWFs) | openEQUELLA (brookes.ac.uk)

[42] EFRAG, 2021. Final Report – Proposals for a relevant and dynamic EU sustainability reporting standard-setting. Download (efrag.org)

Taxonomy[43]. In the present study and adapted from this categorisation, it has been chosen to distinguish three levels of social impacts: impacts on i) individuals (e.g. workers), ii) groups (e.g. affected communities, sectors) and iii) society as a whole. The topics listed above can be linked to these categories in the way set out in Table 1.

Table 1. *Stakeholder categories affected by social impacts*

Social impacts	Categories affected
Distribution of the costs and gains	Individuals, groups
Employment, adequate working conditions and well-being (including burden and stress) of workers involved in sectors and activities that are connected to the sea	Individuals, groups
Employment, adequate working conditions and well-being (including burden and stress) of experts involved in the development and implementation of policies[44]	Individuals, groups
Social impacts linked to the state of the marine environment and/or to the consequences of climate change on coastal communities	Individuals, groups, society
Access to fisheries resources/livelihoods	Individuals, groups

Different types of social impacts can be considered when focusing on marine policies:

- **Distribution of the costs and gains[45]** of the policy/measure between the different groups of stakeholders/sectors/regions/countries – Are efforts fairly shared, costs and gains equally distributed? Are socially fragile groups particularly impacted? Is distribution of access to and benefits from marine ecosystem services equitable?

  Evidence shows that environmental governance generates a variety of benefits and costs which tend to be distributed heterogeneously among local stakeholders. The pursuit of sustainability, in both social and environmental terms, is affected by the (real and perceived) fairness of the costs and benefits' distribution[46]. Haas *et al.* (2022)[47] identified umbrella and specific actions to move towards a more sustainable ocean governance, in particular in response to the risk *"inequitable or unfair distribution of access to and benefits from marine*

[43] Platform on Sustainable Finance, 2022. Final Report on Social Taxonomy. Final Report on Social Taxonomy (europa.eu)

[44] While impacts on experts supporting the development and implementation of a policy (e.g. the MSFD) can be considered in an *ex-ante* or *ex-post* assessment of this policy, or in the context of assessments related to employment policies and working conditions, it is unlikely it is considered within the frame of socio-economic analyses required under the given policy.

[45] Social and economic issues are closely intertwined. Economists and social scientists might look at the same object but with different sensitivities. Here, the costs and benefits will not be discussed from an economic perspective ("how many euros?"). Rather, we will look at whether these costs and gains are distributed fairly, or whether burden is passed on to particular actors/sectors, potentially impacting professionals and populations' well-being.

[46] GURNEY G. G., MANGUBHAI S., FOX M., KIATKOSKI KIM M. and AGRAWAL A., 2021. Equity in environmental governance: perceived fairness of distributional justice principles in marine co-management. Environmental Science & Policy, 124, 23-32. doi:10.1016/j.envsci.2021.05.022

[47] HAAS, B., MACKAY, M., NOVAGLIO, C. *et al.*, 2022. The future of ocean governance. Reviews in Fish Biology and Fisheries 32, 253–270. https://doi.org/10.1007/s11160-020-09631-x

ecosystem services". These actions include: fair-decision-making processes; explicit benefit-sharing arrangements; building capacity for those unheard; having a more balanced stakeholder representation; genuine co-production to strengthen voices of all those involved; bottom-up pressure from powerful groups influencing some Member States to drive negotiations.

- **Employment, adequate working conditions and well-being (including burden and stress) of workers**, considering two different sub-categories:
  (1) **People involved in sectors and activities that are connected to the sea** and that may exert pressure on marine ecosystems and be beneficiary of marine ecosystem services. A policy measure can encourage the creation of new jobs in a given sector or region, while affecting negatively job offers in other sectors/regions[48]. For instance, the energy transition may result in new jobs in the (offshore) wind energy industry, but in a decrease in employment in the fossil industry. Particular attention should be given to traditional sectors such as small-scale fisheries, also taking account vulnerable groups such as foreign workers which may be impacted by EU agreements (e.g. SFPAs) and whose working conditions are often below European standards;
  (2) **Experts involved in the implementation of marine policies** (from the public administration, support consultancy, etc.). A policy can affect work organisation components such as work autonomy, level of teamwork and job rotation, pace of work and work intensity[49]. These elements can in turn affect the well-being of workers and the work-life balance. The MSFD review[50] has revealed the MSFD implementation usually comes with extra work for constant human resources within the public administration, thus leading to high work intensity and potential pressure on these human resources.

- **Social impacts linked to the state of the marine environment and/or to the consequences of climate change on coastal communities** (food availability/livelihoods, health, well-being, displacement of populations, relocation of economic activities…). Climate change has a number of direct and indirect impacts on fishers and coastal communities (e.g. declines and shifting distributions in fisheries, rising sea levels, flooding, erosion, increased storm events). Significant attention has been given to adaptation solutions (e.g. infrastructure to protect and stabilise the shoreline, dune rehabilitation, beach nourishment, restoration of mangroves[51]), but these processes can exclude local people from decision-making, therefore increasing the marginalisation of certain groups and undermining local resilience[52]. The state of the marine environment also directly interferes with human health (both physical and mental). The concept of "One Health", interconnecting the health of people, animals and ecosystems, is being more and more used to describe the interactions between human health and ocean

[48] European Commission, 2021. Better Regulation Toolbox. Better regulation: guidelines and toolbox | European Commission (europa.eu) (Last viewed on 08/08/2022)
[49] European Commission, 2021. Better Regulation Toolbox. Better regulation: guidelines and toolbox | European Commission (europa.eu) (Last viewed on 08/08/2022)
[50] Milieu Consulting and ACTeon, 2022 (forthcoming). Support to the implementation of the Marine Strategy Framework Directive.
[51] BONGARTS L. T., REY-VALETTE H., CHAUMILLON É., CAMUS G., ALMAR R., CAZENAVE A., CLAUDET J., ROCLE N., MEUR-FÉREC C., VIARD F., MERCIER D., DUPUY C., MÉNARD F., ROSSEL B.A., MULLINEAUX L., SICRE M.-A., ZIVIAN A., GAILL F. and EUZEN A., 2021. Designing Coastal Adaptation Strategies to Tackle Sea Level Rise. Front. Mar. Sci. 8:740602. doi: 10.3389/fmars.2021.740602
[52] BENNETT, N.J., 2018. Navigating a just and inclusive path towards sustainable oceans. Marine Policy. https://doi.org/10.1016/j.marpol.2018.06.001

health. There are more and more research connecting the ocean with social sciences, e.g. psychology.

Box 1. Ocean and Human Health: How coastal environments benefit our mental well-being[53]

The framework of "Ocean and Human Health" is an interdisciplinary field investigating the interactions between the marine environment and humans. It aims at promoting the sustainable use of the ocean and its benefits for human health. Research shows that the emotional experience of the sea can influence the well-being of coastal and inland residents. The coast contributes to experiencing positive emotions and help cope with daily stressors. The ocean is beneficial for mental well-being, however it is unclear why and how.

- **Access to fisheries resources/livelihoods** for coastal communities – If the policy or measure threatens the access to marine space and resources of coastal communities traditionally relying on marine areas for their livelihoods, potential compensation should be considered. Some MPAs are compared to a form of "ocean grabbing", which is defined as *"the dispossession or appropriation of use, control or access to ocean space or resources from prior resource users, rights holders or inhabitants […] through inappropriate governance processes and might employ acts that undermine human security or livelihoods or produce impacts that impair social–ecological well-being"*[54]. Some fisheries allocation decisions and management practices have also been contested as they disregard the rights and livelihoods of small-scale fishers and local communities[55]. Compensation can be a way to reduce the impact for affected communities.

2.2.2. Measures and actions that help capturing or dealing with social issues

- **Analyses and assessments** carried out to assess social impacts and challenges related to proposed measures. Different types of analyses and assessments can be carried out to assess social issues, including those designed in the first place to deal with economic issues, such as Cost-Benefit Analyses (CBA), Cost-Effectiveness Analyses (CEA) or Multi-Criteria Analyses (MCA). Social issues can also be addressed in Social Impact Assessment, in assessment of ecosystem services or in specific studies, focusing for instance on social sustainability in the marine field.

Box 2. Addressing social vulnerability in CBA for flood risk management[56]

Most traditional CBAs used for flood risk management focus on avoided asset damages without really accounting for the interests of the (relatively) poor. Poor people are often more vulnerable and exposed to floods. Traditional CBAs will generate investment in areas where people are already relatively better off, leaving the

[53] This work, conducted by Marine SEVERIN (Flanders Marine Institute – marine.severin@vliz.be) was presented during the EU workshop Refreshing H20 policy – or (full title) Refreshing the roles social and economic thinking, methods and tools can play to support EU Blue policy transition delivering sustainability, resilience and security - took place from January 30th to February 1st, 2023, in Rotterdam, the Netherlands.

[54] BENNETT N.J., GOVAN H., SATTERFIELD T., 2015. Ocean grabbing. Marine Policy, Volume 57, 61–68, http://dx.doi.org/10.1016/j.marpol.2015.03.026.

[55] BENNETT, N.J., 2018. Navigating a just and inclusive path towards sustainable oceans. Marine Policy. https://doi.org/10.1016/j.marpol.2018.06.001

[56] This work, conducted by Jarl KIND (De Waterwerkers – the Netherlands, jarl.kind@dewaterwerkers.nl) was presented during the EU workshop Refreshing H20 policy – or (full title) Refreshing the roles social and economic thinking, methods and tools can play to support EU Blue policy transition delivering sustainability, resilience and security - took place from January 30th to February 1st, 2023 in Rotterdam, the Netherlands.

poor exposed. These CBAs therefore raise questions about fairness and social justice of flood risk management policies, as they do not consider "social vulnerability" or "socio-economic resilience" of individuals, communities or societies. A novel framework that integrates social vulnerability and socio-economic resilience into "flood" CBAs has been produced. It distinguishes between collective flood risk, individual recoverable flood risk, and individual catastrophic flood risk, leading on different recommendations on what to do and whom to target.

- **Measures directly linked to social aspects** (e.g. supporting measures dedicated to vulnerable groups in the PoMs, specific indicators…) and **consideration of the "social" in measures that do not primarily target it**. Where negative social impacts of a measure or policy have been identified, solutions may exist to mitigate them. There could be for instance an exception for the most disproportionally affected stakeholders (e.g. vulnerable groups) or accompanying measures intending to mitigate negative effects, such as longer implementation periods, training and job search measures to cope with job loss. Some EU funds can be mobilised for this purpose (e.g. European Social Fund Plus – ESF+ and European Globalisation Fund – EGF)[57]. The social implications of a measure should be considered even when the measure targets other objects (e.g. species, biodiversity) in the first place. The strong environment-centred focus of marine conservation policies should not make forget that humans are part of the ecosystems as well. Cultural and traditional settings must for instance be considered when implementing measures, as some communities historically, traditionally or culturally rely on certain marine areas for their livelihood. The restricted access to fisheries resources or space for some communities (e.g. Territorial User Rights for Fisheries (TURFs)[58]) can be an efficient measure to sustain existing livelihoods while ensuring the conservation of marine resources[59]. If the loss of jobs and/or livelihoods cannot be avoided, it should be mitigated to the extent possible. When appropriate, potential compensation by supporting measures can also be considered to minimise the negative impacts on affected communities.

- **Capacity building/education/training**, **awareness** of ocean issues[60] and **change in behaviour**. Further communication and awareness-raising around ocean issues, in particular targeting an audience less aware of them, would contribute to foster an inclusive dialogue. Ocean literacy initiatives have their role to play in this respect, and can contribute to make a change in behaviour towards more sustainable practices.

[57] European Commission, 2021. Better Regulation Toolbox. Better regulation: guidelines and toolbox | European Commission (europa.eu) (Last viewed on 08/08/2022)

[58] *"Area-based fishing rights, commonly referred to as Territorial Use Rights for Fishing programs, or TURFs, allocate secure, exclusive privileges to fish in a specified area to groups, or in rare cases individuals."* https://fisherysolutionscenter.edf.org/catch-share-basics/turfs (Last viewed on 12/08/2022)

[59] GRIMMEL, H., CALADO, H., FONSECA, C. and SUAREZ DE VIVERO, J. L.,2019. Integration of the social dimension into marine spatial planning – Theoretical aspects and recommendations. Ocean & Coastal Management, 173, 139–147. doi:10.1016/j.ocecoaman.2019.02.013

[60] These social aspects can also be part of the category on "governance and decision-making processes"; the three blocks defined are not impermeable, with possible overlaps.

Box 3. Playing on the emotional level: a way towards behaviour change[61]

There are many ways to try to change behaviour and practices, both among professionals and the wider public. One of the levers is to act on the emotions of individuals. Innovative approaches at the interface between science and art can be used to serve this purpose. Using art, design and tools that appeal to the senses can strike an emotional chord and gently invite behavioural change. It is a different way to communicate and raise awareness of environmental issues. This type of approach can easily be adapted to the target audience. In particular, it can be effective with children who use emotional rather than rational mechanisms. This type of approach makes it possible to reach a diverse audience, for example people who speak different languages.

- **Funding conditionalities** as a lever for fostering sustainable social conditions. The introduction of funding conditionalities could be an effective lever for the creation of sustainable social conditions in the marine field, e.g. within the frame of the EU Taxonomy. The EU Taxonomy is a common classification system established under the Taxonomy Regulation[62] in June 2020 that provides criteria defining sustainable economic activities[63]. The EU Taxonomy can be used by companies, investors and policymakers to identify economic activities that can be considered environmentally sustainable for sustainable investment purposes. The system initially focused on the environmental pillar of the sustainability, even though economic activities had to meet minimum social safeguards (in particular UN Guiding Principles (UNGPs) and Organisation for Economic Cooperation and Development (OECD) guidelines on multinationals) to be recognised as "green" (Article 18). In early 2022, the Platform on Sustainable Finance, an EU advisory body, published its Final Report on Social Taxonomy[64] aiming at advising the EC on potential extensions of the taxonomy framework beyond environmental objectives and defining clearly what can be considered a "social investment". The report covers topics such as decent jobs, improved living standards, better healthcare and creating more inclusive and sustainable communities[65]. The report is based on international norms and principles[66]. The recommendations should feed into an EC report on describing the provisions that would be required to extend the scope of the Taxonomy Regulation as set under Article 26 – this should have already been done by 31 December 2021. Despite the advice report provided by the Platform, the introduction of a "social taxonomy" has been put on hold at least until the end of this parliamentary term in 2024, in particular due to "lack of political will" and "absence of international standards to measure social impact"[67]. While

[61] This illustration box is inspired by the work conducted by Juan P. VELÁSQUEZ (Linnaeus University - Sweden, juan.velasquez@lnu.se) and that was presented during the EU workshop Refreshing H20 policy – or (full title) Refreshing the roles social and economic thinking, methods and tools can play to support EU Blue policy transition delivering sustainability, resilience and security - took place from January 30th to February 1st, 2023 in Rotterdam, the Netherlands.

[62] Regulation (EU) 2020/852 of the European Parliament and of the Council of 18 June 2020 on the establishment of a framework to facilitate sustainable investment, and amending Regulation (EU) 2019/2088. EUR-Lex - 32020R0852 - EN - EUR-Lex (europa.eu)

[63] EU taxonomy for sustainable activities | European Commission (europa.eu) (Last viewed on 10/08/2022)

[64] Platform on Sustainable Finance, 2022. Final Report on Social Taxonomy. Final Report on Social Taxonomy (europa.eu)

[65] See Annex 1 for further information.

[66] Universal Declaration of Human Rights; International Covenant on Economic, Social and Cultural Rights; International Covenant on Civil and Political Rights; ILO Declaration on Fundamental Principles and Rights at Work; ILO Tripartite Declaration of Principles concerning Multinational Enterprises and Social Policy; European Convention on Human Rights; European Social Charter; Charter of Fundamental Rights of the European Union; European pillar of social rights; SDGs; UNGPs; UN Global Compact; OECD guidelines for MNEs.

[67] Social taxonomy could prove a step too far for EU | Impact Investor (impact-investor.com) (Last viewed on 10/08/2022)

environmental objectives and criteria can be science-based, social impact and progress would be harder to quantify. If new regulations are introduced for EU companies, the question also arises as to whether the goods and services entering the EU market comply with these regulations.

Nevertheless, there are other existing EU initiatives that may facilitate sustainable social investment at the EU level, such as the EU's Corporate Sustainability Reporting Directive[68] and the EU's Social Economy Action Plan[69]. The former, also known as the Non-Financial Reporting Directive[70], requires large companies to report on the social (and environmental) impacts of their activities. Meanwhile, the Social Economic Action Plan was launched in December 2021 to coordinate financial support for the social economy which covers entities sharing the principles of *"the primacy of people as well as social and/or environmental purpose over profit, the reinvestment of most of the profits and surpluses to carry out activities in the interest of members/users ("collective interest") or society at large ("general interest") and democratic and/or participatory governance"*[71].

2.2.3. Governance and decision-making processes

Stakeholder and wider public engagement plays a crucial role in social acceptance. It can take several levels of inclusion in policy decisions, ranging from providing information to active involvement in decision-making processes. These levels can be summarised in three approaches[72]:

- **Awareness raising**, **information/knowledge provision**;
- **Consultation**, a limited form of stakeholder and wider public feedback into decision-making processes;
- **Empowerment and participative decision-making approaches**, a more participatory form of engagement giving greater control to participants and enabling them to play key roles in decision-making.

Beyond these three categories, the scale of stakeholder and wider public engagement can be seen as a continuum from full top-down approaches to bottom-up ones, with initiatives of co-management where stakeholders are empowered to share decision-making power[73].

[68] Corporate sustainability reporting | European Commission (europa.eu) (Last viewed on 10/08/2022)
[69] European Commission, 2021. Building an economy that works for people: an action plan for the social economy. EUR-Lex - 52021DC0778 - EN - EUR-Lex (europa.eu)
[70] Directive 2014/95/EU of the European Parliament and of the Council of 22 October 2014 amending Directive 2013/34/EU as regards disclosure of non-financial and diversity information by certain large undertakings and groups. EUR-Lex - 32014L0095 - EN - EUR-Lex (europa.eu)
[71] European Commission, 2021. Building an economy that works for people: an action plan for the social economy. EUR-Lex - 52021DC0778 - EN - EUR-Lex (europa.eu)
[72] AITKEN, M., HAGGETT, C., RUDOLPH, D., 2014. Wind Farms Community Engagement Good Practice Review. https://core.ac.uk/download/pdf/84000113.pdf
[73] HOGG, K., DI FRANCO, A., CALÒ, A., KRSTINIC, P., SANTAROSSA, L. 2019. A governance Toolkit for managing Small-scale Fisheries in Mediterranean Marine Protected Areas. FISHMPABLUE 2 project – Interreg MED. 52 pp. FishMPABlue 2 - Detail (interreg-med.eu)

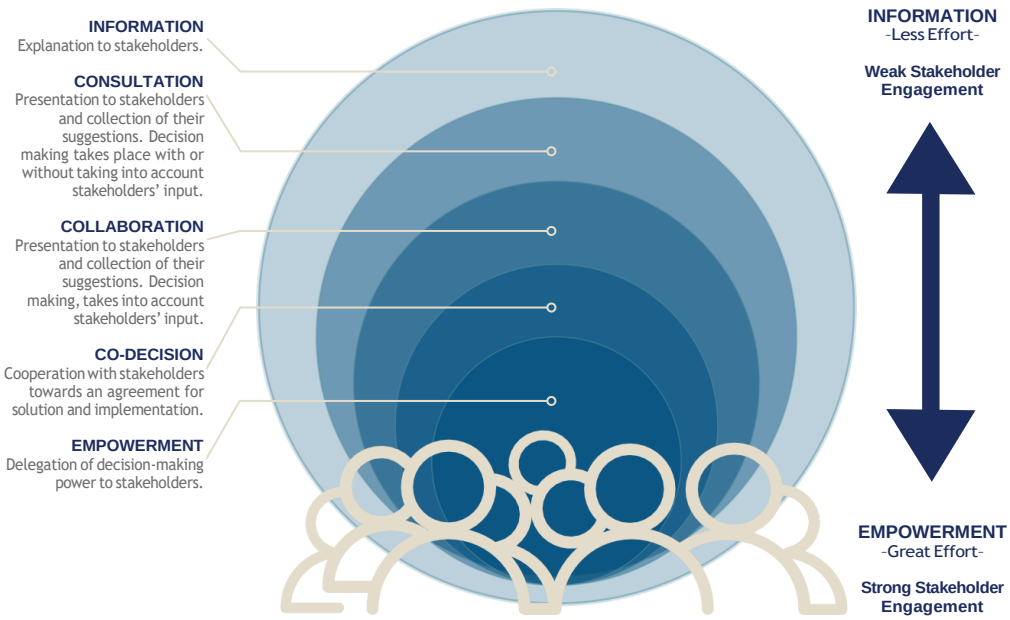

Figure 1. *Different degrees of stakeholder and wider public management*[74]

These different forms of engagement may be relevant in the case of marine policies and strategies. These options are not mutually exclusive and can be combined. Particular attention should be paid to the following issues:

- Do all sectors/groups have equal access to information? Is the knowledge produced being used in the policy-making process?
- Are all sectors/groups impacted directly or indirectly by the policy and its implementation – including vulnerable groups and sectors usually not considered (e.g. local communities) – represented in and have access to decision-making processes?

2.3. Potential place of social issues along the policy cycle

As a way to structure the analysis, social aspects can be studied in relation to the different stages of a policy cycle, as proposed below (Figure 2):

[74] Figure from: HOGG, K., DI FRANCO, A., CALÒ, A., KRSTINIC, P., SANTAROSSA, L. 2019. A governance Toolkit for managing Small-scale Fisheries in Mediterranean Marine Protected Areas. FISHMPABLUE 2 project – Interreg MED. 52 pp. FishMPABlue 2 - Detail (interreg-med.eu)

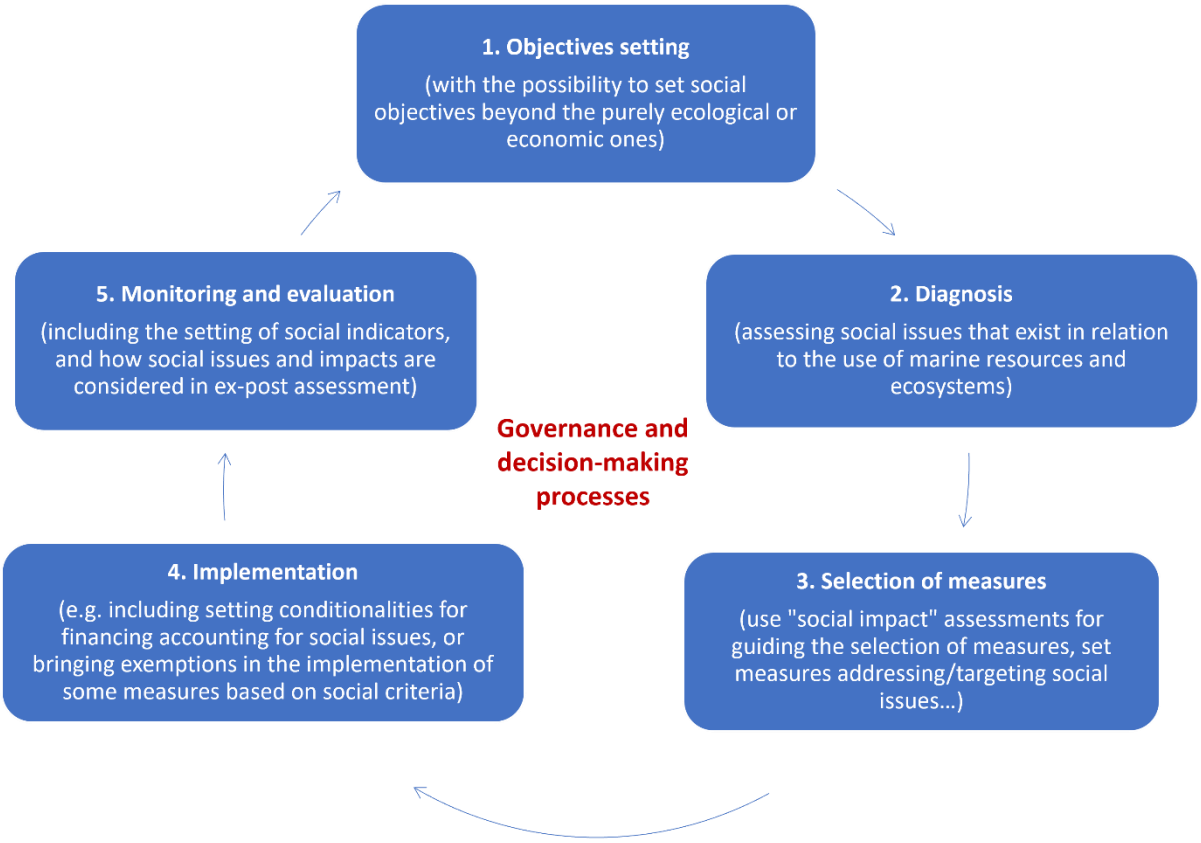

Figure 2. *Potential place of social issues along the policy cycle*

Social issues and challenges are relevant to each stage of the policy cycle, from the setting of objectives of marine policies (defining social objectives) to the setting of social indicators aiming at assessing the social impacts of marine policies after a full implementation cycle. Social issues are also being raised by governance and decision-making processes that occur throughout the entire implementation cycle.

3. Methodological approach

Social issues and challenges have been studied at different decision-making scales considering a broad public intervention framework as far as oceans and coasts are concerned. They have been analysed from the perspective of the implementation of the MSFD in priority. Have also been considered the social issues of parallel key (marine and freshwater) policies and strategies that can be relevant to, and source of inspiration for, the MSFD implementation process, in particular:

- The Water Framework Directive (WFD) in coastal areas;
- The Maritime Spatial Planning Directive (MSPD), addressing the sharing of marine space with social issues relevant to blue economy sectors;
- Blue sector policies and developments, in particular in relation to fisheries (the Common Fisheries Policy (CFP));
- Integrated Coastal Zone Management (ICZM) at the land-sea interface.

Social aspects have been studied from the global to local scales, taking into account relevant international/regional/sea basin marine-related initiatives (e.g. Regional Sea Conventions (RSCs)) as well as existing local instruments, projects and measures (Marine Protected Areas (MPAs), offshore windfarm projects…). The study provides examples mainly from EU Member States, but also from the UK, and to a lesser extent from other countries internationally when relevant examples could be found.

The analysis mobilised both literature review and interviews. Relevant information and data on social issues were extracted from several sources:

- Documents produced as part of the implementation process (formal reporting and accompanying studies, in relation to the ESA, Programmes of Measures (PoM), etc.);
- *Ex-ante* and *ex-post* assessment studies - or regulations and policies – of major projects when these were publicly available;
- Articles and reports from research activities and projects.

Semi-structured interviews were conducted with selected researchers, experts (MSFD experts, social scientists working on the protection of marine resources…), key government/public service representatives and key (blue economy) stakeholder representatives, civil society, also accounting for representatives at lower scales (coastal areas, MPAs, offshore windfarm projects…) (Annex 2).

The report provides a review of the literature available, presents case studies covering a diversity of situations and topics, and highlights key messages. Each case study has been presented in a common case study template, further described in part 5 of this report. The report also includes a chapter with lessons and possible recommendations as a source of inspiration when considering social issues and challenges within the frame of the MSFD implementation in particular.

4. State of play of the consideration of social aspects in marine policies and strategies

4.1. International scale

On a global level, the United Nations (UN) adopted in 2015 the 2030 Agenda for Sustainable Development which brings together 17 **Sustainable Development Goals (SDG)** established by UN Member States and to be achieved by 2030. SDG 14 is dedicated to life below water and aims to *"conserve and sustainably use the oceans, seas and marine resources for sustainable development"*[75]. This SDG mainly focuses on the state of marine ecosystems and therefore on the environmental pillar of sustainability, while pointing to the importance of fair access and benefit sharing in relation to marine resources (targets 14.7 and 14.b). Social (and economic) issues are primarily addressed through other SDGs such as SDG 5 on gender equality, SDG 8 on decent work and economic growth and SDG 10 on reduced inequalities, without further link with the blue economy or oceans in general. The achievement of a sustainable ocean is often narrowed to its ecological dimension, concealing economic and mostly social issues which are not addressed to the same extent.

Employment issues (in particular labour rights and working conditions) are usually the first components of social sustainably being addressed when it comes to the marine field, as shown by the main institutional frameworks addressing social issues in blue economy sectors[76]. Many ocean-related policies and management practices are often guided only by evidence from the natural sciences, the human dimensions being given less attention[77].

Social norms (e.g. rights, justice, equity) and **good governance principles** (e.g. participation, transparency, accountability) are often handled in guidance documents and/or enshrined in international legislation[78]. For instance, Aichi Target 11 of the Convention on Biological Diversity (CBD) states that MPAs should be both "effectively and equitably managed"[79]. Guidance related to its implementation suggests that *"communities […] should equitably share in the benefits arising from protected areas and should not bear inequitable costs"*[80].

There are also numerous international guidance documents and policies addressing **social standards in fisheries** (e.g. human rights, well-being, livelihoods, tenure, food security, working welfare, participation). Those include the FAO Voluntary Guidelines for Securing Sustainable Small-Scale

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[75] Goal 14 | Department of Economic and Social Affairs (un.org) (Last viewed on 10/08/2022)

[76] International Labour Organisation, 2006. Maritime Labour Convention. wcms_763684.pdf (ilo.org)
International Labour Organisation, 2007. Work in Fishing Convention. Convention C188 - Work in Fishing Convention, 2007 (No. 188) (ilo.org)

[77] BENNETT, N.J., 2018. Navigating a just and inclusive path towards sustainable oceans. Marine Policy. https://doi.org/10.1016/j.marpol.2018.06.001

[78] BENNETT, N.J., 2018. Navigating a just and inclusive path towards sustainable oceans. Marine Policy. https://doi.org/10.1016/j.marpol.2018.06.001

[79] Convention on Biological Diversity, 2010. Convention on Biological Diversity (CBD) decision X/2. The Strategic Plan for Biodiversity 2011–2020 and the Aichi Biodiversity Targets. Convention on Biological Diversity, Nagoya, Japan. Aichi Biodiversity Targets (cbd.int) (Last viewed on 25/08/2022)

[80] Convention on Biological Diversity, 2011. Strategic Plan for Biodiversity 2011–2020. Further information related to the technical rationale for the Aichi Biodiversity Targets, including potential indicators and milestones. Convention on Biological Diversity, Nagoya, Japan. https://www.cbd.int/doc/meetings/cop/cop-10/information/cop-10-inf-12-rev1-en.pdf.

Fisheries, FAO Voluntary Guidelines on the Responsible Governance of Tenure, FAO Code of Conduct for Responsible Fisheries, and the ILO Work in Fishing Convention.

The **High-Level Panel for a Sustainable Ocean Economy**[81], also known as the Ocean Panel, is working on **oceans' social sustainability** as well. The Ocean Panel is a global initiative launched in 2018 that gathers 17 world leaders with the aim to sustainably manage 100 % of the ocean area under national jurisdictions by 2030. The Ocean Panel has put forward an Ocean Action Agenda with five key areas of transformation, one of them being "ocean equity". The 2030 Outcome expected is as follows: *"People have equitable access to ocean resources, benefits are fairly distributed and the most vulnerable are protected from the risk of harm"*[82]. The Ocean Panel has identified seven priority actions to achieve this outcome[83].

The **Global Ocean Accounts Partnership** is also working on the integration of the social dimension into **ocean accounting**. An ocean account can be described as *"a structured compilation – of consistent and comparable information: maps, data, statistics and indicators – concerning marine and coastal environments, including related social circumstances and economic activity"*[84]. The aim of these accounts is to inform and support public policy decision-making about oceans. Health, poverty and social inclusion could be accounted for in ocean accounting. The System of National Accounts[85] suggests for instance using a Social Accounting Matrix approach to link sub-populations of concern (e.g. women, low-income, self-employed) with economic sectors. Sub-populations of concern could also be extended to include coastal communities, small-scale fishers, seaweed farmers, mangrove harvesters, local villagers, etc. The ecosystem services such as coastal protection and flood protection could also be disaggregated by sub-populations of concern, to evaluate the quantity of services used by these sub-populations relative to their size. It would also be possible to make links between ocean services and health, e.g. in terms of nutrition received from the ocean and recreational benefits.

More generally on water governance, the **OECD** provides 12 principles for governments to design and implement **effective and inclusive water policies**[86]:

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[81] WRI Ocean Panel – High level panel for a sustainable ocean economy (Last viewed on 29/08/2022)

[82] Ocean Equity – WRI Ocean Panel (Last viewed on 29/08/2022)

[83] 1) Require transparent, responsible business practices that engage and benefit coastal communities, including small-scale fishers, and protect the rights of all workers in ocean industries.
2) Create the conditions to facilitate the full engagement of women in ocean activities to help unlock their economic and social potential and empower them to safeguard naturel resources while enhancing opportunities to access decent work.
3) Recognise and respect the interests of coastal communities and rights of Indigenous Peoples and implement policies that require consideration of the particular importance of marine resources for these groups.
4) Create inclusive governance by incorporating indigenous and local knowledge and interests, particularly those of women and youth, in planning and decision-making processes.
5) Promote integrity across ocean governance and ocean industries, enforce transparency and accountability in public service and public finance and take robust action against corruption.
6) Enhance domestic revenue administration through modernised, progressive tax systems, improved tax policy and more efficient tax collection.
7) Promote international cooperation to combat child labour and forced labour and eliminate trafficking in persons and contraband along supply chains in the ocean economy.

[84] Global Ocean Accounts Partnership, 2021. Technical Guidance on Ocean Accounting for Sustainable Development.

[85] The System of National Accounts (SNA) is the international statistical standard that countries use to measure the economy.

[86] The OECD Principles on Water Governance - OECD (Last viewed on 20/03/2023)

- Principle 1. Clearly allocate and distinguish roles and responsibilities for water policymaking, policy implementation, operational management and regulation, and foster co-ordination across these responsible authorities;
- Principle 2. Manage water at the appropriate scale(s) within integrated basin governance systems to reflect local conditions, and foster co-ordination between the different scales;
- Principle 3. Encourage policy coherence through effective cross-sectoral co-ordination, especially between policies for water and the environment, health, energy, agriculture, industry, spatial planning and land use;
- Principle 4. Adapt the level of capacity of responsible authorities to the complexity of water challenges to be met, and to the set of competencies required to carry out their duties;
- Principle 5. Produce, update, and share timely, consistent, comparable and policy-relevant water and water-related data and information, and use it to guide, assess and improve water policy;
- Principle 6. Ensure that governance arrangements help mobilise water finance and allocate financial resources in an efficient, transparent and timely manner;
- Principle 7. Ensure that sound water management regulatory frameworks are effectively implemented and enforced in pursuit of the public interest;
- Principle 8. Promote the adoption and implementation of innovative water governance practices across responsible authorities, levels of government and relevant stakeholders;
- Principle 9. Mainstream integrity and transparency practices across water policies, water institutions and water governance frameworks for greater accountability and trust in decision-making;
- Principle 10. Promote stakeholder engagement for informed and outcome-oriented contributions to water policy design and implementation;
- Principle 11. Encourage water governance frameworks that help manage trade-offs across water users, rural and urban areas, and generations;
- Principle 12. Promote regular monitoring and evaluation of water policy and governance where appropriate, share the results with the public and make adjustments when needed.

4.2. EU scale[87]

4.2.1. MSFD

Article 8.1.c of the MSFD asks for an **Economic and Social Analysis (ESA)** of the use of marine waters, but does not prescribe how this analysis should be performed. The reporting guidance[88] on the MSFD ESA provides the following definition for ESA: *"A socio-economic analysis aims to identify the impact on human welfare of a given policy. This includes economic as well as social aspects, and may include consideration of the distribution of these impacts across stakeholders. In light of this definition, an explicit distinction between 'economic' and 'social' analysis is not necessary"*. The guidance describes two different approaches for the analysis, namely the ecosystem services approach and the marine water accounts approach.

In line with what has been agreed upon in the guidance document, many Member States have limited the social component of the MSFD ESA to assessments of employment in different maritime sectors.

[87] This section also provides the declination of the MSFD at the national level.
[88] European Commission, 2018. Economic and social analysis for the initial assessment for the Marine Strategy Framework Directive. DG Environment, Brussels. pp 66 (MSFD Guidance Document 1).

Sometimes assessments of (some) cultural ecosystem services have been included. Very little attention is paid to affordability aspects, impacts on (vulnerable) groups or mechanisms to ensure that these groups receive a share of benefits that can be expected from the MSFD implementation and improvements in the ecological status of marine ecosystems[89].

When Member States have considered social issues in their socio-economic assessments, they have considered in particular[90]:

- *Activity dynamics, geographic distribution and employment* (France);
- *Effects on employment and prices, different benefits of measures including different non-economic values of improvements to marine environment* (Germany);
- *Employment in sectors using the sea and distribution/scale of the benefits of the marine ecosystem services* (Latvia).

Social issues have received some attention in different assessments, in particular the analysis of the social impact expected by the adoption of new measures of the PoM, cost-benefit analysis of measures, ESA of the use of marine waters and marine ecosystem services assessment. When available, the **PoMs of some Member States** for the second cycle of the MSFD implementation have been analysed.

- **The Netherlands** (NL) did not take any social assessment of the measures. Nevertheless, it presented different measures corresponding to social process and that will be undertaken in the implementation of the PoM. These measures are mainly related to **education and raising awareness**, in particular regarding the litter issue. NL launched for instance the "Clean beaches programme" that focuses on the exchange of knowledge and the support for cooperation projects. NL also developed a measure aiming at raising awareness of the litter issue among land-based and water managers in order to increase the acceptance of structural measures in management areas. NL also launched the "Waste at School" programme targeting primary and secondary schools to raise awareness about the litter challenge. Awareness-raising activities targeting fishermen are also organised to promote environment-friendly practices. For fishermen undergoing training in fishing schools, a four-day course "Fish with Future" has been developed. National experience has led NL to promote sustainable fisheries education in the OSPAR area. In 2019, on the initiative of NL, OSPAR adopted a recommendation on the reduction of marine litter through sustainability reduction programmes for fishermen.
- **Belgium** also introduced measures mainly related to **education and awareness-raising**, e.g. raising awareness about waste management in marinas, promoting knowledge and environmental awareness in maritime training, implementing consultation and awareness-raising on measures within shipping to reduce underwater noise effects on cetaceans.
- **Finland** (FI) developed **awareness-raising** measures as well, e.g. guidance to contribute to recreational litter reduction in coastal areas, information campaign to reduce underwater noise caused by boating, development of good practices to prevent nutrient discharges in cooperation with port operators… FI also intends to **engage stakeholders in environmental measures** to increase their ownership, e.g. by carrying out participatory monitoring and

[89] Milieu Consulting and ACTeon, 2022 (forthcoming). Support to the implementation of the Marine Strategy Framework Directive – EQ 5: "To what extent is the distribution of the MSFD costs and benefits considered adequate?"

[90] ACTeon, 2021. Assessing current practice with the application of socio-economic assessments to support marine policy/MSFD implementation.

evaluation to assess and monitor the effectiveness of sea basin plans. FI also has an extensive communication-focused measure that aims to raise awareness of marine management objectives in general, among *inter alia* citizens, operators and public authorities. This measure also contributes to indicate how consumers themselves can influence the state of the sea and its biodiversity. The FI PoM also includes a qualitative social assessment of the measures which primarily assesses the effects on health, living conditions and comfort. The implementation of the PoM is expected to have positive indirect effects on these three aspects, with the wider impacts due to a combined effect of the measures. For instance, measures to reduce eutrophication and litter will have a positive impact on comfort and health, including through the reduction of toxic algal blooms and the cleanliness of beaches. The establishment of MPAs is considered as a potential contributor to the principle of equality by guaranteeing an equal access to the protected sites.

- The social-related measures of **Romania** (RO) are also mainly related to **capacity building and awareness-raising**. RO proposed for instance a measure consisting of developing training and awareness programmes on risks, legal issues, best practices and management actions to prevent and mitigate the impact of non-indigenous species. Together with Bulgaria, RO developed a measure to support coordinated establishment of awareness-raising campaigns targeting businesses (e.g. commercial agents, beach operators, fishermen) and the public (e.g. tourists, students, children) related to the sources and consequences of marine litter on the environment and the need for waste recycling. Both countries also jointly developed educational campaigns and provided support for advising local fisheries groups on the effective use of environment-friendly techniques and equipment.
- **Sweden** (SE) implemented measures related to **capacity building and education**, e.g. providing different types of guidance targeting different audiences (e.g. guidance on the content of management documents for MPAs). SE also supports initiatives to promote, organise and carry out beach cleaning in particularly affected areas. SE investigated socio-economic impacts related to the implementation of the PoM, in particular the distributional effects of the new measures.
- In **Ireland** (IE), socially related measures are linked to **public consultation and participation**, e.g. organising a series of workshops with key stakeholders on the further development of measures. IE also carries out **awareness raising and education** activities targeting the public and marine professionals. In order to reduce illegal dumping, IE uses for instance a multi-faceted approach which includes enforcement, reporting, public awareness and education through the Anti-Dumping Initiative. IE supports initiatives which promote public pride in local amenity/natural beauty or reinforce social norms against littering. IE takes part in the Clean Coasts Programme, working with communities to help protect marine and coastal ecosystems. Activities include beach cleaning, site-specific citizen science, and public awareness raising (e.g. highlighting the impacts of plastics on wildlife with local artists). The Blue Flag Beach and Green Coast Award programmes aim to raise awareness to tackle the sources of litter. IE also continues to promote and support schools' education programme, including the themes on litter and waste, biodiversity, and global citizenship marine environment. IE also launched the Aquaculture Information Management System, an online mapping viewer of licensed aquaculture sites aiming to **improve access to information**.
- **Poland** (PL) also implements measures related to **education and information**, such as educational campaign to better protect Marine Natura 2000 sites or educating aquarists and anglers on the dangers of releasing specimens of invasive alien species into the natural environment. PL also supports voluntary work to clean riverbanks and beaches by lakes.

- **Germany** (DE) develops **information campaigns**, e.g. raising awareness of the impact of the UV filters in sunscreen on the marine environment, and other activities aiming to **provide information** to the public and school classes (e.g. in relation to sustainable fisheries). The PoM makes specific reference to the potential **social impacts of measures**, i.e. impacts on human health and social well-being, which includes for instance healthy working conditions, human safety, cultural and social needs.

Overall, the socially related measures implemented by Member States are mainly linked to education, awareness raising, information and consultation.

As part of their **Initial Assessment**, some countries (e.g. the Netherlands, Sweden, Finland, Latvia) performed a Contingent Valuation Method (CVM) study on the Willingness To Pay (WTP) for achieving Good Ecological Status (GES) to better understand the public's relationship with the ocean and to assess whether citizens think it is important to achieve GES and if so, why. This type of analysis is useful to better understand wider society's perceptions about the marine environment and evaluate to what extent citizens support marine protection measures.

In addition, the distribution of the MSFD costs and benefits and the extent to which they are considered adequate have been studied as part of the MSFD review[91]. Most of the costs associated with the MSFD have been borne by the authorities and government services in charge of the MSFD implementation so far. However, this statement must be nuanced and differences can be observed depending on the country. It may also be difficult to know how much stakeholders actually spend to comply with the MSFD requirements. Economic sectors have so far faced limited adjustment costs, but some sectors such as fisheries, recreation and tourism, aquaculture, offshore energy production, industry, agriculture and ports and harbours will have to bear costs to implement new measures and comply with the MSFD requirements in the future. The benefits delivered are limited for the time being, both for government services and for the economic sectors. There are still many uncertainties in the assessment of existing costs and benefits which makes it difficult to assess the fairness of their distribution. The distribution of costs should in particular be fair in relation to the Polluter Pays Principle mentioned in the preamble of the MSFD, stating that *"environmental damage should, as a priority, be rectified at source and that the polluter should pay"*[92]. As part of the online public consultation carried out under the MSFD evaluation, respondents were asked whether resources invested by polluters to implement the MSFD (in line with the Polluter Pays Principle) were sufficient; 90 % of respondents answered that the resources were "not sufficient". Feedback from focus groups with Member States, Regional Sea Conventions, scientific institutes and NGOs were in the same vein[93].

[91] Milieu Consulting and ACTeon, 2022 (forthcoming). Support to the implementation of the Marine Strategy Framework Directive – EQ 5: "To what extent is the distribution of the MSFD costs and benefits considered adequate?"
[92] Marine Strategy Framework Directive, 2008. EUR-Lex - 32008L0056 - EN - EUR-Lex (europa.eu)
[93] Online Public Consultation – Question 39

4.2.2. MSPD

By definition[94], Maritime Spatial Planning (MSP) is a process that aims to achieve ecological, economic and social objectives. This definition is supported in the 2014 EU MSP Directive[95] which stipulates that through endorsing EU's 2020 strategy for smart, sustainable and inclusive growth, as well as the Integrated Maritime Policy (IMP), Member States shall take into account environmental, economic, social (such as delivering high levels of employment, productivity and social cohesion, for example) and safety aspects.

These social aspects, and more, can be taken in consideration in various steps of the MSP implementation and planning cycle (preparation, scenario analysis and outcome)[96]. But perspectives of social justice and equity should be addressed during all stages of planning. In particular, there is a reference to recognition, representation and distribution[97],[98].

Since the launching of the Directive, several studies revolving around social sustainability and social inclusion in MSP reveal that the MSP can indeed contribute to the improvement of many social issues (such as poverty, or gender considerations for example) through the following actions[99]: stakeholder engagement; public participation (i.e. indigenous communities); developing policies with principles of social equity and inclusion; integrating social information in spatial data (culturally sensitive areas, traditional tenure systems, belief systems and knowledge…); spatial planning with social considerations (i.e. allocation of space to certain uses, prohibition of using the marine space through establishing MPAs…); improving social acceptance of specific marine sectors (i.e. offshore renewable energy; coastal and offshore aquaculture); addressing social impacts/side effects of marine sectors and MSP on different actors; social learning and socio-ecological innovation; planning for climate change (and understanding how climate change does not affect all places uniformly, thus considering differences in socio-ecological vulnerabilities). The MSP-GREEN project[100], funded by the European Maritime, Fisheries and Aquaculture Fund (EMFAF) and aiming to see how maritime spatial plans can contribute to the ambition of the European Green Deal, is also taking social issues on board.

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[94] Marine Spatial Planning (MSP) is a public process of analysing and allocating the spatial and temporal distribution of human activities in marine areas to achieve ecological, economic and social objectives that have been specified through a political process. (UNESCO IOC, 2009. Marine Spatial Planning A step by step approach towards Ecosystem Based Management)

[95] https://www.eea.europa.eu/policy-documents/directive-2014-89-eu-maritime/

[96] Swedish Agency for Marine and Water Management. 2021. Poverty and gender considerations in Marine Spatial Planning: Conceptual and analytical framework. https://www.havochvatten.se/download/18.4d67699c177b5eb7753a2207/1614617420303/msp-poverty-genderframework-working-paper-feb-2021.pdf

[97] BENNETT, N. J., CISNEROS-MONTEMAYOR, A. M., BLYTHE, J., SILVER, J. J., SINGH, G., ANDREWS, N., CALÒ, A., CHRISTIE, P., DI FRANCO, A., FINKBEINER, E. M., GELCICH, S., GUIDETTI, P., HARPER, S., HOTTE, N., KITTINGER, J. N., LE BILLON, P., LISTER, J., DE LA LAMA, R. L., MCKINLEY, E., SCHOLTENS, J., SOLÅS, A. M., SOWMAN, M., TALLONI-ÁLVAREZ, N., TEH, L. C. L., VOYER, M. and SUMAILA, U. R., 2019. Towards a sustainable and equitable blue economy. Nature Sustainability, Vol. 2, pp. 991-993. https://doi.org/10.1038/s41893-019-0404-1

[98] SAUNDERS, F., GILEK, M., IKAUNIECE, A., TAFON, R. M., GEE, K. and ZAUCHA, J., 2020. Theorizing Social Sustainability and Justice in Marine Spatial Planning: Democracy, Diversity, and Equity. Sustainability, Vol. 12, No. 6, 2560. https://doi.org/10.3390/su12062560

[99] UNESCO-IOC, 2021. MSPglobal International Guide on Marine/Maritime Spatial Planning.

[100] MSP-GREEN (mspgreen.eu) (Last viewed on 17/03/2023)

Box 4. The MSP-GREEN project: how are social issues taken on board in maritime spatial plans?

The MSP-GREEN project started in 2023 and will last for two years. An analysis grid has been developed to review maritime spatial plans (e.g. strategic façade documents in France). The socially related criteria of the grid consider: the **inclusion of stakeholders** in the different phases of the process, and **representation** of different stakeholder groups; **gender balance**; the different types of **experts** involved; the inclusion of **local knowledge** and the use of **participatory science**; **access to information**; the capacity of stakeholders to **influence the decision-making process**.
Stakeholder interviews and case studies are also planned in order to identify "good practices" and to formulate recommendations for policy makers so that maritime spatial planning can be better used to achieve the objectives of the European Green Deal.

4.2.3. CFP and EMFAF

The CFP[101] makes several references to the social dimension. In particular, it states that it shall ensure *"fishing and aquaculture activities are environmentally sustainable in the long-term and are managed in a way that is consistent with the objectives of achieving economic, social and employment benefits"* (Article 2). It is also specified that *"before measures are included in the multiannual plans, account shall be taken of their likely economic and social impact"* (Article 9).

It has been recognised that the conservation and environmental issues relating to fisheries are the focus of attention, while the social aspect is very little addressed. There is an increasing awareness that more attention should be given to social aspects in the fisheries sector. In order to get a better picture of the social dimension of fisheries within the EU or EU-flagged vessels, social data are collected every three years by Member States and analysed by the Scientific, Technical and Economic Committee for Fisheries (STECF) Expert Working Group[102]. These social variables are: Employment by gender; Full Time Employment (FTE) by gender; Unpaid labour by gender; Employment by age; Employment by education level; Employment by nationality; Employment by employment status; Total FTE National.

Some priorities of EMFAF, which aims in particular to support the CFP (among others), are directly addressing social issues. It is particularly the case of the priority n°3 *"Enabling a sustainable blue economy in coastal, island and inland areas and fostering the development of fishing and aquaculture communities"*. Under this priority, EMFAF supports the emergence and activities of Fisheries Local Action Groups (FLAGs), a tool to support community-led local development.

4.2.4. WFD[103]

The text of the WFD[104] stipulates the following points:

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[101] Regulation (EU) No 1380/2013 of the European Parliament and of the Council of 11 December 2013 on the Common Fisheries Policy, amending Council Regulations (EC) No 1954/2003 and (EC) No 1224/2009 and repealing Council Regulations (EC) No 2371/2002 and (EC) No 639/2004 and Council Decision 2004/585/EC. EUR-Lex - 02013R1380-20190814 - EN - EUR-Lex (europa.eu)

[102] DOERING, R., FITZPATRICK, M. and GUILLEN GARCIA, J. editor(s), 2020. Scientific, Technical and Economic Committee for Fisheries (STECF) – Social dimension of the CFP (STECF-20-14). EUR 28359 EN, Publications Office of the European Union, Luxembourg, 2020, ISBN 978-92-76-27169-7, doi:10.2760/255978, JRC123058. EN (europa.eu)

[103] The WFD covers both inland and coastal waters, however it is difficult to make a real distinction between the two in this study.

[104] Directive 2000/60/EC of the European Parliament and of the Council of 23 October 2000 establishing a framework for Community action in the field of water policy. EUR-Lex - 32000L0060 - EN - EUR-Lex (europa.eu)

- *"The success of this Directive relies on close cooperation and coherent action at Community, Member State and local level as well as on **information, consultation and involvement of the public, including users**"* (Paragraph 14);
- *"To ensure the **participation** of the general public including users of water in the establishment and updating of river basin management plans, it is necessary to **provide proper information** of planned measures and to report on progress with their implementation with a view to the **involvement** of the general public before final decisions on the necessary measures are adopted"* (Paragraph 46);
- *"Member States may have regard to the **social**, environmental and economic effects of the recovery* (of costs for water services)*"* (Article 9).

The social aspect of sustainability is mainly linked to empowerment of civil society, participation of the general public and specific stakeholders. In order to assist competent authorities with the public participation in relation to the WFD, the Working Group 2.9 Public Participation has produced a guidance document[105].

Box 5. Towards a socially diverse and inclusive water management in the Netherlands[106]

In 2019, the PvdA network of diversity and inclusiveness organised a masterclass on diversity and inclusiveness in water management. Following this masterclass, the regional public water authority Amstel, Gooi and Vecht (one of the 21 regional water authorities in the Netherlands) has developed a proposal on diversity and inclusiveness within the regional water authorities. Regional water authorities were invited to sign a Diversity Charter. By signing the Charter, the regional water authorities will commit to respect multiple dimensions of diversity, including: cultural, ethnic and religious diversity, gender, age and work ability. The Amstel, Gooi and Vecht water authority also tries to involve residents as much as possible and inform them about the regional water authority's work and choices. The regional water authority aims to appeal the widest possible group, ensuring that young and old take an interest in the regional water authority's work.

4.3. Regional scale

In the **Mediterranean Sea basin**, cultural differences between the "North" and the "South – East" Mediterranean result in a large disparity in considerations of social dimensions – namely the gender issue. The six Mediterranean regional activity centers – as part of the **Barcelona Convention'**s institutional structure – have each developed a strategy for gender equality, and the UNEP Mediterranean Action Plan even assigned a gender officer role.
Moreover, and to bridge this gender gap, the MedProgramme[107] – funded by the Global Environment Facility (GEF) – established a Gender Community of Practice in 2021 (see box below).
Additionally, the **Union for the Mediterranean** has launched its intergovernmental monitoring mechanism on gender equality in 2020[108]. While it is not centered on maritime issues, it is a step forward towards addressing gaps, and even just promoting dialogue around this subject.
In regard to marine planning and coastal zone management, the UNEP MAP Priority Actions Programme/Regional Activity Centre (PAP/RAC) considers stakeholder engagement and the inclusion of different social groups as a main priority in their processes. This dimension is even considered in

[105] Public participation in relation to the water framework directive. Guidance document No 8 - Publications Office of the EU (europa.eu) (Last viewed on 11/01/2023)
[106] STOFFEL, S., 2021. Spraakwater – Sociale inclusiviteit - Is een divers en inclusief waterbeheer nodig? Water Governance. 555499 (wur.nl)
[107] https://www.thegef.org/projects-operations/projects/9607 (Last viewed on 17/03/2023)
[108] https://ufmsecretariat.org/gender-equality/ (Last viewed on 17/03/2023)

their online toolkit on Maritime Spatial Planning governance: **The Mediterranean MSP planning and progress toolkit**[109].

✓ Major stakeholders and their interests and influence identified

For e.g. Local populations, social groups and marine-related occupations, individual municipalities and communities, marine-users, businesses and their associations, chambers, cooperatives, unions and networks, voluntary organisations. Infrastructure operators. Organisations in related domains including environment, energy, tourism, health, defence. Representative bodies for the disabled and other excluded groups.

✓ Social actors identified, upholding diversity and gender equity

For e.g. individuals with political or administrative powers, controls and responsibilities including mayors, councillors, regional and national politicians and officials, and their geographic scope. Opinion formers and community leaders in social and media bodies. Individuals with skills and expertise in public administrations, universities, and the voluntary and private sector in marine-and related domains.

✓ Methods and awareness tools to ensure social actors stakeholders' engagement throughout the process identified, designed and deployed

***Figure 3**. Captures from the Mediterranean MSP Planning and Progress Checklist online platform, related to the social consideration of MSP governance (PAP/RAC, 2023)*

Box 6. The MedProgramme Gender Community of Practice[110]

The MedProgramme, funded by GEF, has 10 beneficiary countries from the Euro MED region (non-EU). Its four components reflect the priorities adopted by the contracting parties to the Barcelona Convention (reduction of land-based pollution; enhancing sustainability and climate resilience in the coastal zone, protecting marine biodiversity and knowledge management).

In 2021, and as part of its Gender Mainstreaming Strategy[111], the MedProgramme established the Gender Community of Practice (CoP). Consisting of executing partners and national counterparts, its aim was to create a forum that will help enhance knowledge on gender equality and women empowerment within the context of MedProgramme activities – centred around sustainable development.

The establishment of this CoP was preceded by a survey – a rapid gender capacity assessment – aiming to establish a baseline on the partners' capacity related to gender mainstreaming and revealed a need to have a gender training amongst the partners. So far two thematic Gender CoP meetings took place focused on (1) gender issues in biodiversity and (2) gender issues in chemicals and waste.

Results of these meetings include sharing of experiences, presenting an online database on Gender and Environment courses, implementation progress of capacity building activities, presentation of case studies and best practices from beyond the Mediterranean, with an ultimate goal to have a better understanding of how gender discussions can be translated into actions.

The **Baltic Sea** presents a very different context, where social considerations are farther ahead in the policy level discussions, especially with regards to Maritime Spatial Planning. Besides the national MSP authorities' work being consolidated (and in some cases – evaluated), some initiatives are being put in place to integrate social dimensions in ocean planning. The project "Taking Social Sustainability to the Sea" is an example from the Baltic Sea Region illustrated below.

[109] https://msp.iczmplatform.org/planning-tools/mediterranean-msp-planning-and-progress-checklist/ (Last viewed on 17/03/2023)

[110] https://news.iwlearn.net/the-medprogramme-gender-community-of-practice-genderresponsive-environmental-action-in-the-mediterranean?locale=en%253Fembed%253Dtrue%253Fembed%253Dtruecategories%3Fmore%3Dtrue (Last viewed on 17/03/2023)

[111] https://iwlearn.net/resolveuid/be67692b-dca0-4c70-8ba8-50bd7b9ce3ce (Last viewed on 17/03/2023)

Box 7. Taking social sustainability to the sea: Strengthening the Social Pillar in Marine Spatial Planning[112]

Funded by the Foundation for Baltic and East European Studies, this project took place between June 2019 and February 2022. Four Baltic Sea countries were involved, Germany, Latvia, Poland and Sweden and aimed to answer the question of "what could social sustainability look like in MSP?".
This question entails looking into **social values** related to MSP and associated with the marine environment/uses and activities. And since countries are likely to interpret sustainability and social objectives in different ways, this framework was examined in **four different countries**.

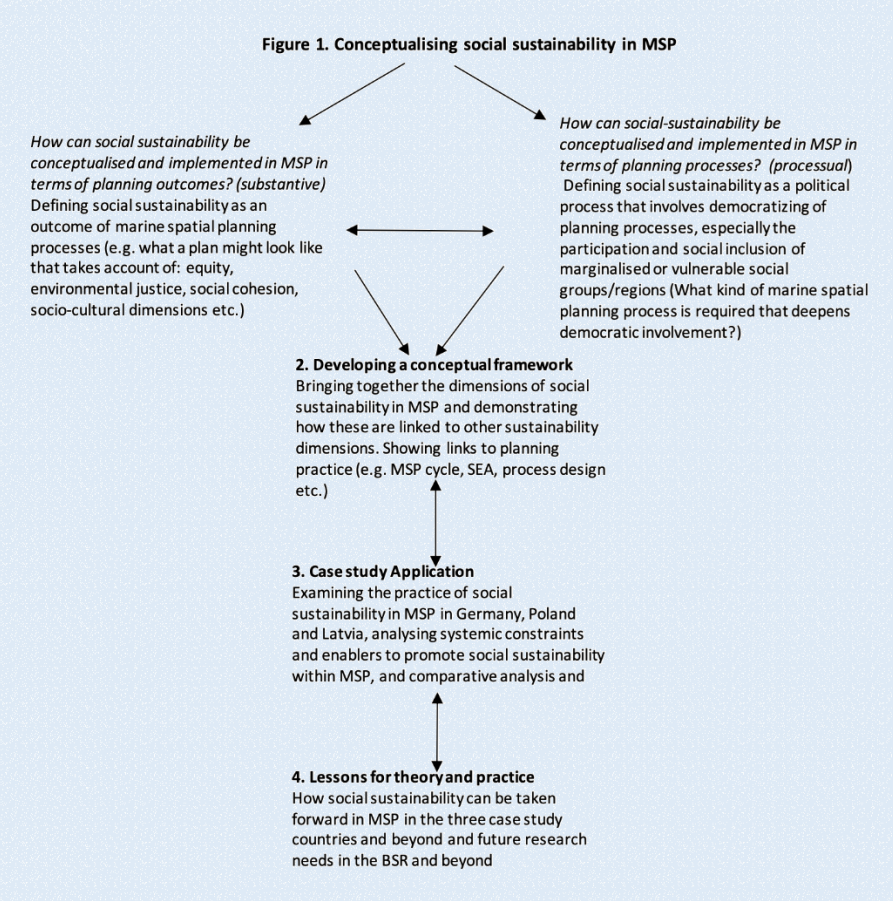


Figure 4*. Conceptualising social sustainability in MSP*

Social sustainability was examined both in (a) the **inputs** and in (b) the **outcomes** of the MSP process:
This means examining (a) the s**ocio-political process** of MSP, looking into issues such as power, democracy, and participation and (b) the **outcomes** of the MSP process, meaning the **benefits** of a socially sustainable marine plan and its contribution to equity, environmental justice, and social cohesion.
The novelty of this approach is bringing together various bodies of knowledge as part of the comprehensive conceptual framework.

At the **Black Sea** level, there have been different European funds directed to encourage the involvement of social experts in marine field during the last years, e.g. within the frame of the following projects: Black Sea Assistance Mechanism[113], H2020 CONNECT BS[114], BRIDGE BS[115], DOORS BS[116]. In

[112] https://maritime-spatial-planning.ec.europa.eu/projects/taking-social-sustainability-sea-strengthening-social-pillar-marine-spatial-planning
[113] Black Sea Assistance Mechanism (BSAM) | Common Maritime Agenda for the Black Sea (europa.eu) (Last viewed on 22/03/2023)
[114] Home | Black Sea Connect (connect2blacksea.org) (Last viewed on 22/03/2023)
[115] BRIDGE Black Sea (Last viewed on 22/03/2023)
[116] DOORS Black Sea (Last viewed on 22/03/2023)

Romania, the incorporation of human dimensions into marine and coastal policy and management is still lacking. The number of Romanian social experts with coastal or marine interests is very small, mainly because of lack of educational programs specialised on the marine social aspects (most of the trainings are for traditional sectors like shipping or tourism), as well as a lack of funds from the national or regional authorities to encourage social experts to study the marine/maritime sectors. Nevertheless, workshops and focus groups with actors are organised to improve their knowledge, capacity development and ocean literacy.

4.4. National & local scales

Integrated Coastal Zone Management (ICZM) is defined by the European Environment Agency as *"a dynamic, multidisciplinary and iterative process to promote sustainable management of coastal zones"*[117]. ICZM assesses the societal goals in a given coastal area and takes measures using informed participation and cooperation of all stakeholders. ICZM pursues a number of objectives, including social objectives, which must be balanced. ICZM relies on the integration of objectives and on the integration of all the instruments needed to achieve these objectives (e.g. policies, sectors, administration). Main objectives of ICZM[118] are related to:

- Protecting people and assets at risk;
- Enhancing sustainability and ecosystem services;
- Economic development of the coastal zone;
- Creating awareness of coastal zone vulnerability and risks, ensuring public information and consultation;
- Good governance, with the creation of a knowledge base.

Actions taken for the implementation of ICZM and the achievement of social-related goals include public participation in coastal zone management, capacity building of ICZM staff and exchange of knowledge and good practices among similar coastal zones.

At the EU level, the European Commission launched the OURCOAST initiative with the aim to share and make accessible European ICZM experiences and practices[119]. This initiative provides in particular good examples of participation in practice, recognising that participatory processes (rather than consultative) can be a large task. All the experiences are available in the multi-lingual Europe-wide ICZM database[120]. Key approaches of ICZM include participation (sharing of information, transparent communication, consensus-building under stakeholders and the general public, informed decision-making), knowledge-based decision-making (using indigenous and local knowledge) and socio-economic benefits.

[117] integrated coastal zone management — European Environment Agency (europa.eu) (Last viewed on 10/01/2023)
[118] Integrated Coastal Zone Management (ICZM) - Coastal Wiki (Last viewed on 11/01/2023)
[119] European Commission, 2010. OURCOAST ICZM in Europe – Integrated Coastal Zone Management – Participation practices in Europe. doi: 10.2779/43519
[120] OURCOAST, the European portal for Integrated Coastal Zone Management — English (europa.eu) (Last viewed on 11/01/2023)

Box 8. Engagement of coastal communities in coastal change adaptation: an example from the UK

In the UK, the government set up the Coastal Change Pathfinder Programme in 2009[121]. Fifteen local councils were commissioned to explore different approaches to adaptation to coastal change through close working within affected communities. The projects focused notably on adaptation planning and community engagement.

As an illustration, the Jurassic Coast Pathfinder project (south-west England) sought to achieve seven objectives:
1) Coastal communities who are **knowledgeable about what coastal change is and how it affects them**;
2) Coastal communities who are able to **debate about coastal change**;
3) Coastal communities who are able to **actively and meaningfully be part of the decision-making process** regarding coastal change;
4) Coastal communities who are well-prepared and equipped when dealing with coastal emergencies now and in the long term;
5) Coastal communities who are supported in the testing and acceptance of practical action to adapt to coastal change;
6) Future generations of coastal communities who are **aware of the coastal changes facing them**, and better prepared to deal with these changes;
7) A spatial planning system which is well-equipped to reconcile the potential conflicts between a sustainable approach to coastal change on the one hand, and onshore/offshore development pressures on the other.

Other local initiatives include the **Fisheries Local Action Groups** (FLAGs)[122], which are partnerships between fisheries actors and other local private and public stakeholders developed under the third priority (Community-Led Local Development) of EMFAF. FLAGs design and implement a local development strategy to address the economic, social and/or environmental local needs. This local strategy is used to select and provide funding to local projects that contribute to achieving the objectives of the strategy[123].

Box 9. Examples of social actions implemented in FLAGs: the case of the FLAG "three estuaries" in France

The FLAG "three estuaries" has set up various projects with a social aim, in particular to **raise awareness**, promote **social inclusion and diversity**, and provide **social support** for fisheries professionals:
- Project of the "House for All", which aims to promote social and intergenerational diversity and offers awareness-raising activities related to the coastal heritage;
- Social support projects for fishermen to cope with increasing administrative constraints, e.g. training in the online declaration of fishing catches.

At the local level, examples of taking the social dimension into account can also be found in **marine protected areas**, whether in terms of governance or measures implemented.

Box 10. Taking the social dimension into account in marine protected areas: the case of the Iroise Marine Natural Park in France

Created in 2007, the Iroise Natural Marine Park (INMP) is the first French natural marine park. Although the consideration of the social dimension is not written in stone in the park management plan, the INMP is still implementing a number of measures that address social issues:
- The INMP's operating model favours structures that generate a lot of jobs, and in particular supports small-scale fishing, benefiting the appeasement and the social acceptance of the INMP by this sector;

[121] SS2_UK_Pathfinder_programme.pdf (coastadapt.com.au)
[122] Other examples of social actions carried out by FLAGs are provided in case study 1 "Inclusion of vulnerable groups in fisheries communities: the example of FARNET" in annex of this report.
[123] FLAG Factsheets | FARNET (europa.eu) (Last viewed on 11/01/2023)

- The INMP's management measures are designed to take into account the capacity of the structures to absorb the socio-economic impact and is based on dialogue with stakeholders;
- The INMP carries out many environmental education and awareness-raising activities, targeting professionals and users, the general public and children. The INMP's measures are intended to be inclusive. For example, the park provides subsidies to schools that do not have the means to set up environmental education activities themselves. In this way, the park seeks to make education accessible to all;
- During its open days, the INMP offers trips to the sea that are also accessible to people with limited means.

The philosophy of the INMP is to promote the **human-nature nexus**: protecting marine ecosystems, while allowing humans to maintain their socio-economic activities in a sustainable manner. It is considered that the exploitation of natural resources is a know-how that is part of the cultural heritage.

Another example tackling social issues on a local level is the CoastWise project, a collaboration between five partner institutions targeting the improvement of the integrated assessment of South Africa's land-sea interface[124].

Box 11. CoastWise: Improving knowledge for integrated management of the land-sea interface in South Africa[125]

In order to cater to the growing needs for a strong foundation for decision making of integrated coastal conservation in South Africa, the CoastWise project developed new data and methods for coastal assessment and planning in support of improved integrated management in South Africa, and by strengthening coastal literacy. This project strongly considers the **cultural significance** of coastal and marine sites by local communities, and takes into account the benefits and values assigned to these ecosystems by them. To achieve this, the project developed a framework for identifying Culturally significant areas in the South African coastal zone, and used innovative digital solutions (such as machine learning and digital outreach tools) and translated them into new spatial layers, fact sheets, video policy briefs targeting ministers and decision-makers… to improve the knowledge base.
The project will provide a strong foundation that coastal managers can build on**, with the ultimate goal of securing and restoring these ecologically and culturally significant spaces.**

At the national level, based on six study sites in the Mediterranean and Brittany, a research project on the socio-economic impacts of offshore wind power ("EOLENMER") has just been launched in France[126].

Box 12. The French EOLENMER project: an observatory of the socio-economic impacts of offshore wind energy

The EOLENMER research project, which has just started in 2023 and will last four years, is studying the socio-economic impacts of offshore wind power in France. This project will particularly focus on:
- **Social impacts**, particularly in terms of the perceived effect of wind farms on the landscape;
- The **counterparts** that can be provided when developing a wind farm. "Counterpart" can refer to environmental compensation, but also financial compensation and financing by energy companies to support some local projects (e.g. creation of a museum, a path on the coastline, etc.). These compensations aim at improving the social acceptability of the wind project;
- The **distribution of the benefits** generated by the wind farm between the different categories of stakeholders, assessing to what extent the redistribution can be judged "fair";
- **Participatory democracy**, and the integration of different stakeholders in the decision-making process;
- The **link between art and science**, to better understand the impacts and issues felt by the stakeholders.

[124] https://meerwissen.org/partnership-projects/coastwise (Last viewed on 17/03/2023)
[125] https://science4ocean.meerwissen.org/project/coastwise/ (Last viewed on 17/03/2023)
[126] Unfortunately, there is no website for this project yet.

5. Investigating social issues in case studies

A **series of seven case studies** have been carried out for more in-depth assessments of social issues in the marine field, analysing how these social issues are considered in practice under diverse contexts. These case studies illustrate the diversity of social issues and challenges that are relevant to the use and protection of marine resources and ecosystems, and to the implementation of the MSFD.

The aim of the case studies is to provide **concrete illustrations** of social issues and challenges, and more importantly how these have been considered and addressed in practice. These very diverse case studies cover contrasting contexts, scales and EU policies and policy instruments addressing marine uses and ecosystems. These case studies are far from representing the range of (scientific and operational) initiatives and projects that address marine-related social issues in Europe. However, these can be seen as source of inspiration on how countries could address these, in particular where marine-related social issues are emerging and have not yet been investigated nor addressed.

The following case studies have been selected for the analysis based on the authors' suggestions and with the inputs of POMESA working group members (Table 2).

Table 2. *Case studies investigated*

Case study	Geographical scale	Policy cycle step	Topic(s) covered
Inclusion of vulnerable groups in fisheries communities: the example of FARNET	EU & national (Finland, Spain, France)	Implementation of measures	Social inclusion, social vulnerability, employment
Involving fishermen in MPA governance: the FISHMPABLUE2 project	National (Croatia, France, Greece, Italy, Slovenia and Spain)	Implementation of measures, governance and decision-making processes	Awareness raising, engagement, participatory governance
Civil society involvement and social acceptability of marine energy projects: the MERIFIC project	EU & national (Denmark)	Implementation of measures, governance and decision-making processes	Civil society involvement, social acceptability
Social issues addressed in valuation studies supporting the MSFD implementation: a comparative study in the Baltic Sea	Regional (Baltic Sea – Finland, Sweden and Latvia)	Diagnosis	Awareness raising, people's perception on challenges around the marine environment, welfare/well-being
Ocean Literacy in the UK: a multidimensional social survey	National (England, Wales and Scotland)	Diagnosis	Awareness raising, education
Women in the sustainable blue economy: contributing to gender equality in maritime sectors	Multiple	Implementation of measures	Gender equality
Assessment of the social impacts of environmental measures taken under the MSFD	National (France)	Selection of measures	Impact of measures at the socio-economic level

Case studies were developed based on the analysis of the available literature complemented in some cases by interviews with relevant experts. Each case study is presented following a common template as illustrated below (Table 3).

Table 3. *Common template used for case studies*

Name (+ iconography presenting the country, scale, step in the planning process addressed)	
Where? What context?	Brief description and context Geographical scale ([international], EU or Europe, regional/sea basin, national, local scale)
What?	Which social issues and challenges were encountered? And have been investigated/have received attention in the case study
Who?	Who were the stakeholders mobilised or target audiences of the activities implemented?
How?	How were these issues considered, investigated and/or resolved in practice? Which resources (information, expertise and capacity, financing…) have been mobilised? And who has been in charge of/contributing to activities?
What lessons?	What have been the main results and added value of taking into account social issues and challenges? To the contrary, what problems remains resulting from not taking them (fully) into account? Which room for improvement? Which lessons and relevance for the MSFD implementation?

Key messages from these case studies are presented in the following section "From findings to recommendations", combined with key messages emerging from the literature review and the interviews conducted. The full case studies are available in Annex 3 of this report.

6. Conclusion – From findings to recommendations

6.1. Main findings and challenges

Social aspects are increasingly in the spotlight and given consideration by political and institutional actors mobilised in the protection of marine ecosystems and the sustainable use of marine resources. They are increasingly the focus of actions and initiatives at different geographical scales, and considered independently or in parallel with economic issues and challenges dimension that might have been the main point of attention including in the context of the MSFD implementation.

Social issues cover a wide range of topics and interpretations depending on the disciplines and policies considered. In the context of the CFP for instance, the social dimension relates mainly to socio-economic and socio-demographic criteria, employment by gender or employment by education levels. In the context of the MSPD, different social objectives are being pursued, in particular high levels of employment and social cohesion. By widening the scope, the social dimension can encompass many aspects, *inter alia* fairness, inclusiveness, social acceptability, well-being and health, but also awareness-raising and transparent (and appropriate) dissemination of information to give everyone the opportunity to take an interest and even participate in political decisions relating to the marine environment.

In the context of the MSFD, as illustrated by the assessments carried out in the context of the second cycle and measures proposed in the second PoM, most attention is given to employment (although investigated mainly through the development of indicators using available statistics – thus with a more economic focus), education and raising-awareness. The limited attention given to other social dimensions can relate to the priority given to environmental status (as the priority objective of the Directive). However, as highlighted by several experts consulted, addressing social issues is essential in relation to the acceptance of proposed (environmentally focused) measures or to the "share of burden" in relation to socially vulnerable groups. The human resource constraints faced by Member States for the implementation of the MSFD can also explain the limited attention to social issues. Indeed, limited resources are allocated to the social and economic aspects of the MSFD with priority given to the economic pillars (e.g. the need for carrying out cost-benefit analyses) for which there are clear reporting requirements. It is unclear if the teams in charge of the implementation of the MSFD include social scientists such as sociologists.

Looking beyond EU policies, many actions implemented in the framework of other initiatives can be looked at as sources of inspiration. For instance, social measures taken in the framework of Fisheries Local Action Groups (FLAGs) include providing training and education to vulnerable groups in fisheries communities and supporting them in the search of economic opportunities. Some measures also intend to better engage local communities and professionals in governance processes, e.g. involving fishers in the surveillance or the monitoring of some MPAs. Establishing Communities of Practice, such as the case of Gender within the MedProgramme, intends to accompany practitioners towards action. Similar initiatives and measures could be integrated into the PoMs of the MSFD as accompanying measures addressing social concerns and challenges.

Interviews and case studies have helped identifying a series of challenges faced when considering social issues and challenges, including:

- I**nsufficient awareness of social issues** by those in charge of the implementation of policies despite them being a growing concern;

- **Limited human resources** for the operational implementation of the requirements of policies, and the absence of social science expertise in the teams in charge of the implementation of policies and strategies addressing marine ecosystems and resources;
- The difficulty to reach **vulnerable groups** as a result of the lack of connections and contacts with these groups making it difficult to account for their specific needs and challenges, as well as developing measures targeted these groups that would facilitate the policy implementation;
- **Administrative barriers** making it difficult to implement socially oriented measures. Rules for funding (e.g. of projects and measures aimed at improving the environmental status of marine ecosystems, or the sustainability of fisheries) can be restrictive, while social initiatives often require flexible and innovative approaches potentially implemented at more local scales than the scales considered in the implementation process of EU Directives such as the MSPD and MSFD;
- The **lack of specific educational programs related to social marine sciences**, as well as a **lack of funds** from the national or regional authorities to encourage social experts to study the marine/maritime sectors;
- The **diversity of social challenges faced locally**, making it difficult to aggregate information on these challenges at the regional, sea basin or national scales.

6.2. Recommendations for the MSFD implementation

There are different ways in which the social dimension could be better taken into account in the MSFD implementation, be it in terms of governance, measurement or analysis. More attention could be given for example to:

- The **engagement of all parties in the MSFD governance**, giving specific attention to, and facilitating, the involvement of vulnerable groups (e.g. small-scale fishers, local coastal communities…) to facilitate the effective implementation of the MSFD implementation. For this engagement to be successful, most vulnerable groups must be provided with appropriate means and conditions to participate. Knowledge should be made accessible to anyone (e.g. in terms of language used, subtitles, audio-visual description…)with the provision of targeted trainings and capacity building facilitating everybody's participation in discussions. To improve the contribution of marine stakeholders and of the wider society in the MSFD implementation, participatory processes could go beyond minimum legal (information and consultation) obligations, building e.g. on more local community-based working groups or participatory storyline development guiding interventions. A gender-balanced governance, a principle rarely considered nor made operational in policy implementation, could also be given some attention. Countries' gender agenda could be internalised in the implementation of the marine and maritime policies, building for instance on collaborations with organisations already addressing gender issues;
- **Citizens' perceptions, understanding and acceptability** of marine ecosystems, links between human activities and the state of ecosystems and changes in practices and behaviour required for achieving good status of marine ecosystems. Citizens' surveys can help better understanding these, including the extent to which citizens support protection measures proposed under the MSFD PoM, or how maritime space can be shared between economic sector developments and the protection of ecosystems. Citizens' survey can also

contribute to more **targeted communication** as well as to designing actions that can support **mobilisation and behaviour change** and be integrated into the PoMs;

- **Proposing and implementing socially oriented measures**, such as measures contributing to further engage stakeholders into the MSFD implementation process (e.g. through monitoring or controlling activities) or limiting (negative) social impacts resulting from some ecosystem protection measures. Such measures could be integrated into the PoMs, or be the focus of dedicated accompanying measures of programmes and strategies with social objectives;
- **Give more explicit considerations to social issues and challenges in assessments** required under the MSFD and the MSPD. In relation to the MSFD, these could be considered when revising the initial assessment of the social and economic importance of activities and ecosystems or when assessing measures for the next planning cycle. Due consideration to social impacts complementing already-practiced environmental and economic impact assessments would well fit within **multicriteria analyses** complementing (or integrating) cost-effectiveness and/or cost-benefit analyses. Social indicators could also be given specific attention in the (*ex-ante* and *ex-post*) review and evaluation of strategies and policies carried out at different (regional, national, European) scales. In particular, social dimensions and issues could receive particular attention in the MSFD revision process.

Activities that could support a more systematic attention to social issues and challenges in the implementation of current policies, or in the shaping up of future marine policies (e.g. in the context of the MSFD review) could include:

- **The adaptation of the reporting** on Economic and Social Analyses to complement current reporting requirements with (a limited number of) elements dedicated to social issues and challenges. This would help capturing: (1) the importance of social issues and challenges related to the protection of marine ecosystems; and (2) how these issues are given due consideration in the MSFD implementation process;
- The development of **specific guidance** on social issues and challenges building on the elements presented in this report complemented by additional experiences and practice from MS available in particular in the grey literature. If considered as having added value, such guidance, or annex to existing guidance, could focus on a specific component of social issues (e.g. assessing social impacts – in relation to the assessments supporting the selection of PoM, or addressing social challenges related to information, consultation and participation). In light of the (human) resources mobilised for the implementation of the MSFD, as well as capacity constraints, adding new requirements would however need to be carefully designed;
- Organise the **exchange and sharing of practical knowledge and experiences** on how to take social issues on board – be it by; (1) organising **trainings** or **workshops** dedicated to specific policy process and target audiences, or attending seminars and conferences addressing social issues organised by others (such as the seminar organised by FARNET on social inclusion). These workshops would provide the opportunity to present successful examples of how the social dimension has been taken into account within the framework of the MSFD and other relevant marine policies at different geographical and institutional levels. They could be organised at the EU or Regional Sea basin scales (within the frame of Regional Sea Conventions); (2) facilitate or better organise the **sharing of practice and conversation** within the POMESA WG and beyond (e.g. using the shared Teams and functionalities); or, (3) contributing to the setting up of a wider (European) **Community of**

Practice at the interface between knowledge providers and policy makers, involving researchers, social science experts supporting strategic and policy processes at different scales, executive agencies, MS representatives responsible for the implementation of marine policies, etc.

Strengthening the consideration given to social issues within the MSFD implementation could also benefit from liaising with initiatives already addressing these issues[127], such as: (1) work on social indicators under the CFP[128] that could contribute to forthcoming updates of the MSFD Economic and Social Analysis (ESA); (2) work on the social dimension of fisheries carried out by the ICES working group on social indicators (WGSOCIAL)[129] relevant to the MSFD implementation[130]; (3) social indicators considered in EU funding conditionalities[131] that can be source of inspiration for indicators considered in the assessments of PoM. While there is a clear interest in shedding light on social issues and challenges in many countries, it may however be already too late to consider additional social issues in the current ESA update already underway. For the assessment of the next PoMs, building synergies with other policies and initiatives addressing social issues (CFP, EU funding conditionalities, ICES working groups – see above) could strengthen the attention given to social challenges in the MSFD implementation. Giving greater importance to multicriteria analyses (as compared to Cost-Benefit Analysis limited to monetary values) for supporting the selection of measures and the design of the PoMs could also provide greater opportunities in considering social impacts often addressed with more qualitative evidence.

[127] From discussions that took place at the 28th meeting of the POMESA Working Group in April 2023

[128] See report produced by the Scientific, Technical and Economic Committee for Fisheries (STECF) Expert Working Group on the social dimension of the CFP: EN (europa.eu). To be noted that a new project was launched end of 2022 by EU Social Partners to improve the social dimension of the CFP (see Project launched to to improve the social dimension of the CFP (thefishingdaily.com)).

[129] WGSOCIAL (ices.dk) (Last viewed on 17/05/2023)

[130] ICES, 2021. Working Group on Social Indicators (WGSOCIAL; outputs from 2020 meeting). ICES Scientific Reports. 3:8. 29 pp. https://doi.org/10.17895/ices.pub.7690. Working Group on Social Indicators (WGSOCIAL) (figshare.com)

[131] See for example the recent inclusion of social conditionalities in the allocation of Common Agricultural Policy subsidies (Key reforms in the new CAP (europa.eu) – Last viewed on 01/06/2023). On May 4th 2023, the two responsible European Parliament committees for the financial rules adopted the proposal made by the Commission which includes a social conditionality (PR_COD_1recastingam (europa.eu)) – *"In addition, they propose that the principle of social conditionality be applied horizontally to EU funding, following its inclusion in the new CAP 2021-27, and that concise and proportionate performance indicators are introduced to monitor the impact of Union spending on gender equality"*.

Annexes

Annex 1. Social aspects as defined by main (UN and EU) official and guidance documents

Reference	Social aspects considered	
UN Declaration on the Rights of Indigenous Peoples[132]		- Humans rights and fundamental freedoms - Non-discrimination and equality - Respect for cultural differences - Self-determination - Free, prior and informed consent and access to fair compensation - Right to participate in the "life of the State" - Right to the lands, territories and resources traditionally owned, occupied or used - Effective and appropriate consultation
UN Social Development Goals[133]	Goal 3. Good health and well-being	Ensure healthy lives and promote well-being for all at all ages
	Goal 4. Quality education	Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
	Goal 5. Gender equality	Achieve gender equality and empower all women and girls
	Goal 8. Decent work and economic growth	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all
	Goal 10. Reduced inequalities	Reduce inequality within and among countries
	Goal 16. Peace, justice and strong institutions	Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
European Convention on Human Rights[134]		- Right to life - Prohibition of torture - Prohibition of slavery and forced labour - Right to liberty and security - Right to a fair trial - No punishment without law - Right to respect for private and family life - Freedom of thought, conscience and religion - Freedom of expression - Freedom of assembly and association - Right to marry - Right to an effective remedy - Prohibition of discrimination

[132] United Nations Declaration on the Rights of Indigenous Peoples | Division for Inclusive Social Development (DISD) (Last viewed on 20/03/2023)
[133] Home | Sustainable Development (un.org) (Last viewed on 20/03/2023)
[134] European Court of Human Rights, 2021. European Convention on Human Rights. European Convention on Human Rights (coe.int)

European Social Charter[135]		- Employment - Housing - Health - Education - Social protection - Welfare - Protection of vulnerable persons (elderly people, children, people with disabilities and migrants) - Non-discrimination
EU Charter on Fundamental Rights[136]	Title I. Dignity	- Human dignity - Right to life - Right to the integrity of the person - Prohibition of torture and inhuman or degrading treatment of punishment - Prohibition of slavery and forced labour
	Title II. Freedoms	- Right to liberty and security - Respect for private and family life - Protection of personal data - Right to marry and right to found a family - Freedom of thought, conscience and religion - Freedom of expression and information - Freedom of assembly and of association - Freedom of the arts and sciences - Right to education - Freedom to choose an occupation and right to engage in work - Freedom to conduct a business - Right to property - Right to asylum - Protection in the event of removal, expulsion or extradition
	Title III. Equality	- Equality before the law - Non-discrimination - Cultural, religious and linguistic diversity - Equality between women and men - The rights of the child - The rights of the elderly - Integration of persons with disabilities
	Title IV. Solidarity	- Workers' right to information and consultation within the undertaking - Right of collective bargaining and action

[135] The European Social Charter - Social Rights (coe.int) (Last viewed on 20/03/2023)
[136] EU Charter of Fundamental Rights (europa.eu) (Last viewed on 20/03/2023)

		- Right of access to placement services - Protection in the event of unjustified dismissal - Fair and just working conditions - Prohibition of child labour and protection of young people at work - Family and professional life - Social security and social assistance - Health care - Access to services of general economic interest - Environmental protection - Consumer protection
	Title V. Citizens' rights	- Right to vote and to stand as a candidate at elections to the European Parliament - Right to vote and to stand as a candidate at municipal elections - Right to good administration - Right of access to documents - European Ombudsman - Right to petition - Freedom of movement and of residence - Diplomatic and consular protection
	Title VI. Justice	- Right to an effective remedy and to a fair trial - Presumption of innocence and right of defence - Principles of legality and proportionality of criminal offences and penalties - Right not to be tried or punished twice in criminal proceedings for the same criminal offence
European Pillar of Social Rights[137]	Chapter I: Equal opportunities and access to the labour market	1. Education, training and life-long learning 2. Gender equality 3. Equal opportunities 4. Active support to employment
	Chapter II: Fair working conditions	5. Secure and adaptable employment 6. Wages 7. Information about employment conditions and protection in case of dismissals 8. Social dialogue and involvement of workers 9. Work-life balance 10. Healthy, safe and well-adapted work environment and data protection
	Chapter III: Social protection and inclusion	11. Childcare and support to children 12. Social protection 13. Unemployment benefits 14. Minimum income 15. Old age income and pensions 16. Health care

[137] The European Pillar of Social Rights in 20 principles | European Commission (europa.eu) (Last viewed on 11/08/2022)

		17. Inclusion of people with disabilities 18. Long-term care 19. Housing and assistance for the homeless 20. Access to essential services
Better Regulation Toolbox[138]	TOOL #29. Fundamental rights, including the promotion of equality	- Rights, freedoms and principles as defined by the EU Charter on Fundamental Rights[139]: Dignity, Freedoms, Equality, Solidarity, Citizens Rights and Justice - Non-discrimination and equality
	TOOL #30. Employment, working conditions, income distribution, social protection and inclusion	- Employment - Working conditions - Income distribution, social protection and inclusion, including impacts on perspectives and life-situations of people in, or at risk of poverty
	TOOL #31. Education and training, culture and youth (ETCY)	- Education and training - Culture - Youth
	TOOL #32. Health impacts	- Direct health impacts - Indirect health impacts
Sustainable Finance Platform report on social taxonomy[140]	Decent work (including value-chain workers)	- Promoting decent work - Promoting equality and non-discrimination at work - Ensuring respect for the human rights and workers' rights of affected workers in the value chain
	Adequate living standards and wellbeing for end-users	- Ensuring health and safe products and services - Designing products to be durable and repairable - Providing for cybersecurity and the protection of personal data and privacy - Engaging in responsible marketing practices - Ensuring access to quality healthcare products and services including care services - Improving access to healthy and highly nutritious food - Improving access to good-quality drinking water - Improving access to good-quality housing - Improving access to education and lifelong learning
	Inclusive and sustainable communities and societies	- Promoting equality and inclusive growth - Supporting sustainable livelihoods and land rights - Ensuring respect for the human rights and affected communities

[138] European Commission, 2021. Better Regulation Toolbox. Better regulation: guidelines and toolbox | European Commission (europa.eu) (Last viewed on 08/08/2022)
[139] Charter of Fundamental Rights of the European Union. EUR-Lex - C2012/326/02 - EN - EUR-Lex (europa.eu)
[140] Platform on Sustainable Finance, 2022. Final Report on Social Taxonomy. Final Report on Social Taxonomy (europa.eu)

Annex 2. List of interviewees

Category of stakeholder	Scale	Organisation	Name	Role/function
Expert	EU	FARNET/FAMENET	Urszula BUDZICH TABOR	Senior Local Development expert
Expert	Regional (Black Sea)	Ovidius University of Constanta, Romania	Natasa VAIDIANU	Associate Professor in Maritime Spatial Planning, Environmental Geography, Coastal management
Expert	National (Latvia)	AKTiiVS Ltd.	Kristine PAKALNIETE	MSFD expert involved in ESA, POMESA member
Expert	National (Finland)	Finnish Environment Institute (Syke)	Tommi TIKKANEN	Researcher involved in the MSFD implementation
Expert	National (France)	IFREMER	Adeline BAS	Marine economist involved in the MSFD ESA
Expert/Government representative	International / national (Germany)	Helmholtz Zentrum Hereon – German MSP Authority for the EEZ	Kira GEE	Scientist and researcher at HZH; member of the MSP Authority
Expert/Government representative	National (UK)	Department for Environment, Food and Rural Affairs (UK government)	Mark ATKINSON	Head of Marine Social Science, social and behavioural scientist
Intergovernmental organisation representative	Regional (MED)	UNEP MAP PAP/RAC	Marina MARKOVIC & Veronique EVERS	Programme officers
Civil society representative	National (France)	Women for Sea	Bertille LEFÈVRE	Project manager
Marine natural park representative	Local (France)	Iroise Marine Natural Park	Philippe LE NILIOT	Assistant director
(Blue economy) stakeholder representative	Local (France)	Fisheries Local Action Group “3 estuaries”	Typhaine JAOUEN	FLAG animator

Annex 3. Case studies investigated

Inclusion of vulnerable groups in fisheries communities: the example of FARNET **Scale:** **Policy cycle step**: Implementation of measures	
Where? What context?	FARNET was the network of people and organisations implementing Community-Led Local Development (CLLD) for two successive funding periods, 2007-2013 (as Axis 4 of the European Fisheries Fund) and the 2014-2020 period (as Union Priority 4 of the European Maritime and Fisheries Fund (EMFF)), community led development being the third priority of this fund. Since 2022, with the start of the European Maritime, Fisheries and Aquaculture Fund (EMFAF), FARNET has been replaced by FAMENET which continues support to local development as one of three pillars of its work. FARNET/FAMENET is a technical assistance service that provides capacity building and recommendations of practices to actors involved in the implementation of CLLD, notably to Fisheries Local Action Groups (FLAGs) that develop activities and initiatives contributing to the sustainable development of fisheries, other blue economy sectors and coastal areas[141]. FLAGs are an EU tool operating at local level and using EU Funding programmed and implemented at the national and sub-national scales. FARNET and FLAGs are inclusive mechanisms and processes by nature, supporting local communities to address issues they face while contributing to local territorial development. FLAGs have different objectives addressing the challenges faced by fisheries and coastal communities such as improving local economic development and supporting the development of new services and income opportunities. Responding to the needs identified by the local community, they do not solely focus on social issues, but some of their actions have a strong social component addressing e.g. social inclusion in fisheries communities and the mobilisation of vulnerable groups (e.g. women, disabled people, migrants and refugees). This case study illustrates concrete examples of actions[142] implemented by FLAGs in several EU countries (i.e. Finland, Spain and France) that aim at integrating vulnerable these groups into the fisheries communities and enabling them to benefit from the wider blue economy[143]. Many FLAGs include social organisations as members of their partnership, with budgets allocated to projects dealing with social inclusion that can reach 10 to 30 % of the FLAGs total budget.
What?	**Social inclusion** can be understood as the process of improving the integration of individuals and groups in society[144]. Some individuals or groups, referred to as "vulnerable groups", are particularly at risk of being excluded from actions, decisions and (financial and marine) resource allocation The actions carried out by some FLAGs particularly intend to include and engage vulnerable groups into the fisheries sector, contributing to improving their role in the local community. This approach seeks to gather and build on the contributions of all, including those that are

[141] Welcome to FARNET: the European Fisheries Areas Network | FARNET (europa.eu) (Last viewed on 17/11/2022)
[142] FARNET, 2017. Social inclusion for vibrant fishing communities. FARNET Guide #13: Social inclusion for vibrant fishing communities | FARNET (europa.eu)
[143] These actions were pointed out during the FARNET social inclusion seminar that took place in March 2017. FARNET organises seminars twice a year. There was only one seminar specifically focused on social issues over the past 12 years. However, several other seminars covered topics highly relevant for social inclusion (for example the seminars on Circular Economy or Smart Coastal Areas).
[144] http://www.worldbank.org/en/topic/socialdevelopment/brief/social-inclusion (Last viewed on 28/11/2022)
Joint report by the Commission and the Council on social inclusion, 2004. https://ec.europa.eu/employment_social/soc-prot/soc-incl/final_joint_inclusion_report_2003_en.pdf

traditionally less consulted. Other social issues addressed in this case study include **capacity building, social well-being, gender and cultural heritage.**

Fishing and aquaculture communities are facing increasing challenges related to landing obligation, the drive towards Maximum Sustainable Yield (MSY) as required by the Common Fisheries Policy (CFP), regionalisation and increased competition, seasonality of the activities, generational renewal… The development of complementary means of livelihoods helps addressing the resulting income reduction.

Example of action 1 – Engaging fishers with young generation

In Finland, the Ostrobothnia FLAG[145] supported the "Blue Care project"[146] (2017-2020), a project inspired by the concept of Green Care, which consists of offering health related services linked to nature and the (rural) environment. Greencare projects were first implemented in rural areas to provide alternative or complementary activities to farmers. The Blue Care project aimed at combating marginalisation through fisheries. It particularly focused on **school children facing learning difficulties and low motivation.** It engaged with young people to inspire them for potential careers that could meet the labour needs of the fisheries sector . A programme for 13 to 15-year-old pupils with learning difficulties was designed by participating fishermen and three local schools with the coordination of the Natural Resources Institute Finland (Luke). Learning activities took place once a week at the harbour and on-board fishing boats and were designed to fit the school curriculum by offering application examples in maths, biology and entrepreneurship. Pupils also participated in fishermen's land-based activities (e.g. landing fish, taking care of nets). Fishermen received a financial compensation in return. Similar initiatives could also be developed for older citizens "disconnected from the labour market", e.g. long-term unemployed, workers recovering from burn out… Overall, the Blue Care project promoted **social well-being** as well as **cultural heritage**.

Example of action 2 – Providing women with training to seize new activities' opportunities

In Spain, the Costa da Morte FLAG[147] has developed a training course focused on:

- Providing women working in shellfish gathering and nets repairing (both being seasonal activities) with the required knowledge to develop activities linked with tourism;
- Raising the confidence of women and their interest in such activities;
- Supporting the development of new activities.

A total of 20 women participated to the first training, 13 of them being then involved in tourism related activities. This training boosted women's self-confidence and helped them to better grasp their local role. Beyond trainees, it also attracted the attention of women interested in developing similar projects.

[145] Ostrobothnia FLAG | FARNET (europa.eu) (Last viewed on 28/11/2022)
[146] Blue Care: combating marginalisation through fisheries | FARNET (europa.eu) (Last viewed on 28/11/2022)
[147] Costa da Morte FLAG | FARNET (europa.eu) (Last viewed on 28/11/2022)

	Example of action 3 – Creating innovative jobs for the long-term unemployed and people with disabilities FLAGs from France (Basque Coast FLAG[8]) and Spain (Mariña-Ortegal FLAG[9]) have supported projects aiming at reducing waste from the fishing sector while creating jobs for long-term unemployed and disabled persons. In France, the project[148] aimed at recycling old fishing gear, offering new job opportunities for fishers and for long-term unemployed and disabled persons in disassembling used nets and other gears. This project benefited from the support of the ADELI Association, a FLAG partner with experience in providing sheltered employment to socially excluded people. Training to disassemble and sort gear was provided by fishers. In Spain, an association of disabled persons initiated a project[149] (2013-2014) to reuse plastic from the fisheries sector using sustainable recycling techniques. While having a positive environmental impact, the project provided employment opportunities for people with labour market integration problems. The FLAG funded a practical training course for participants to have the necessary skills in plastic recycling.
Who?	Stakeholders affected are primarily vulnerable groups and individuals usually less consulted (e.g. women, the disabled, the long-term unemployed, the young).
How?	FLAGs activities related to social inclusion are partly funded by EMFAF. The projects conducted by FLAGs involve a variety of partners, some of them working more specifically on social issues, e.g.: • Associations working with socially excluded people (e.g. ADELI association in France); • Education stakeholders; • Municipalities and social services; • Psychologists and physiotherapists. While the main target groups of FARNET are FLAGs and managing authorities, they also call on social scientists and experts when organising events or carrying out studies. Although the activities on the ground are carried out by the FLAGs or their beneficiaries, EU-level support from FARNET provides capacity building for FLAGs to identify and address social issues, as well as inspiring good practices and opportunities for networking and mutual learning between FLAGs from different Member States and regions.
What lessons?	Addressing inclusion issues in relation to the management of fisheries has contributed to:

[148] Création d'un atelier expérimental pour la revalorisation de matériels de pêche usagés et l'insertion professionnelle - DLAL FEAMP (Last viewed on 28/11/2022)
[149] A social enterprise to service the fish auctions | FARNET (europa.eu) (Last viewed on 28/11/2022)

- Improving the contribution and engagement of vulnerable people, thereby increasing their social inclusion and well-being;
- Developing complementarity activities for some blue economy sectors, increasing links with other local stakeholders and citizens;
- Increasing the visibility of women in the fishing sector and supporting their professional development with new skills.

The examples show the importance of mobilising stakeholders from beyond the traditional "marine" communities, in particular stakeholders from the social domain addressing the challenges faced by vulnerable groups.

Insights from the inside: lessons from a FARNET/FAMENET representative

The work conducted under FARNET has contributed to better awareness on social aspects. The information on social issues produced by FARNET has been disseminated to the majority of FLAGs via a seminar on social inclusion and guidance (available in different languages). The FARNET network also provided opportunities for FLAG's managing bodies to learn about measures that can better address social issues. For example, the guidance produced and showcased by FARNET has contributed to increased awareness on gender issues across Europe. And it is hoped that good practices and measures highlighted by FARNET will help FLAGs to develop new projects addressing gender and other social issues. FARNET network is only here to provide guidance; CLLD is a bottom-up approach, and FLAGs managers that are on the ground are the ones to decide how to spend funds. Managing authorities at national level also have their say on the priorities and conditions of support. FARNET network organises two meetings per year with managing authorities. Recently, these meetings have mainly focused on blue economy as a new topic under the EMFAF, but social aspects could be brought again once the blue economy topic "has been digested". Due to the transition the 2014-2020 funding period (coming to an end in 2023) and the new 2021-2027 funding period in which most FLAGs will not be operational before 2024, few new projects will be launched for the time being.

Several constraints and difficulties can be encountered when it comes to implementing actions for social purposes:

- There is still **insufficient awareness of social issues**, in particular as much attention is given to economic and environmental aspects. FLAGs managing organisations and members should be continuously reminded of the importance of social and societal effects, and of the importance of being inclusive;
- **Vulnerable groups are often not very active and can be sometimes difficult to reach**. It is difficult to reach out people at risk of exclusion, as well as young generations in particular in remote areas. The lack of contact with these groups makes it difficult to design support in line with their needs. Minority groups may also struggle to participate as they often lack the education and knowledge needed, or cannot access information (e.g. good practices) when it is in English. In cases these are in place, civil society organisations from the social sector can help reaching vulnerable groups;
- **Administrative barriers** can make it difficult to implement socially oriented measures. Rules for funding are very restrictive, while social initiatives often require flexible and innovative approaches. The rules are sometimes not possible to deal with by small local actors, due to very restrictive conditions. At the EU and national level, the rules are designed for a seven-year period; even if the local situation changes, the rules remain the same until the end of the funding period.

Lessons for the MSFD implementation…

Initiatives similar to the one presented above could be integrated under the Programmes of Measures of the MSFD as accompanying measures addressing social concerns and challenges, including in the search for complementary economic opportunities, training/capacity building and education.

Involving fishermen in MPA governance: the FISHMPABLUE2 project **Scale**: Mediterranean Sea **Policy cycle step**: Implementation of measures, governance and decision-making processes	
Where? What context?	Marine Protected Areas (MPAs) are among the main management tools designed to protect the marine environment. Attempts are being made worldwide to establish a global network of MPAs, towards the objective of protecting 30 % of the ocean by 2030. In the EU, the EU 2030 Biodiversity Strategy sets the target of legally protecting a minimum of 30 % of the EU sea area. At the international scale, delegates agreed to protect 30 % of the world's marine areas by 2030 during the last UN Biodiversity Conference (COP 15). Despite their potential benefits, MPAs may not be accepted by local communities and stakeholders, a situation that can hinder their effectiveness. The application of good governance principles, in particular in relation to participatory governance, can contribute to fully engaging local communities and stakeholders in the MPA management process, potentially contributing to greater social acceptance of management principles and activities. In this context, the FishMPABlue2 project[150] (2016-2019) tested a set of governance mechanisms identified as key to the successful management of small-scale fisheries, within and around MPAs located in the Mediterranean. Its ecological, economic and social impacts were also assessed. As part of this project, a "Governance toolkit"[151] was produced and tested in 11 MPAs from 6 Mediterranean countries (Croatia, France, Greece, Italy, Slovenia and Spain). This test phase made it possible to assess the social acceptance of management measures by stakeholders, in particular fishers. The Governance toolkit intends to support shifts towards co-management of MPAs, where the management responsibility is shared between stakeholders.
What?	Management tools tested in the FishMPABlue2 project belonged to five main areas: • Involvement in decision-making; • Enforcement strengthening; • Knowledge and ownership; • Improvement of small-scale fisheries environmental sustainability; • Improvement of small-scale fisheries profitability. Four tools with a clear social component were designed to improve fishermen's engagement, knowledge and ownership of MPAs, addressing in particular **social acceptance, participatory governance, co-management and awareness-raising**.

[150] FishMPABlue 2 (interreg-med.eu) (Last viewed on 29/11/2022)

[151] HOGG, K., DI FRANCO, A., CALÒ, A., KRSTINIC, P., SANTAROSSA, L. 2019. A governance Toolkit for managing Small-scale Fisheries in Mediterranean Marine Protected Areas. FISHMPABLUE 2 project – Interreg MED. 52 pp. FishMPABlue 2 - Detail (interreg-med.eu)

Tool 1 – Create a permanent and formal cooperation platform to engage fishers in decision-making

The organisation of a cooperation platform was tested in 11 MPAs that had experienced already some sort of fishers' mobilisation prior to the FishMPABlue2 project. The cooperation platform named "Local Governance Cluster" (LGC) brought fishers' engagement one step further in decision-making: the LGC is a formalised joint committee composed mainly of MPA managing bodies and local fishers' representatives, responsible for the main decisions concerning the implementation of the project. In some cases, this was the first time fishers were actively engaged in decision-making. Where appropriate, other stakeholders were incorporated in the LGC, including researchers, local NGOs and representatives of other blue economy sectors.

In 8 of the 11 MPAs, the LGC was taken a step further, with the implementation of dedicated governance mechanisms increasing fishers' engagement and facilitating two-way dialogues. In one MPA, efforts were made to strengthen an existing FLAG, with for example: actions strengthening the capacity of fishers and of their representatives; support to the FLAG in the application of relevant funds; support to fishers to become members of small-scale fisheries organisations (e.g. the Low Impact Fishers of Europe – LIFE network).

Tool 2 – Increase surveillance through direct fishers' involvement

Limited MPA enforcement capacities as well as poorly perceived rules can lead to a high level of non-compliance. Enforcement was identified as a major issue for 10 out of 11 MPAs that considered in the FishMPABlue2 project. To strengthen enforcement and enhance users' compliance with regulations, increased surveillance through fishers' direct involvement was proposed in 6 MPAs. Involving fishers in patrolling contributed to increase the overall surveillance of MPAs while generating an **increased sense of empowerment, ownership and awareness**. Fishers faced difficulties linked to peer-to-peer enforcement. Training courses were provided to fishers interested in surveillance activities.

Tool 3 – Engaging fishers in monitoring activities

Stakeholders' engagement in monitoring and knowledge sharing is a key ingredient of good governance. Giving consideration to local interests and perceptions of resource users in how natural resources could be managed is key to social acceptability. In 5 MPAs, local fishers were directly involved in monitoring activities with differences inn objectives of monitoring and the degree of fishers' involvement. In some cases, fishers monitored pre-defined indicators, and in others, fishers were involved in determining what additional monitoring could be done. Interested fishers were provided with adequate training to monitoring fundamentals (e.g. principles of data collection process, monitoring forms, how to report data…) as well as with the equipment required to carry out monitoring. As part of the FishMPABlue2 project, the Egadi MPA (Italy) also

	developed a participatory process for designing a voluntary "Code of Conduct" that includes guidance for MPAs monitoring. This Code is in line with the FAO "Code of Conduct for Responsible Fishing". ***Tool 4 – Raising the awareness of fishers, MPA managers and the local community*** Within the FishMPABlue2 project, different strategies were employed to raise the awareness on marine ecosystems of MPA managers, governing bodies, marine stakeholders and the wider community, as a means to increase buy-in and support to management activities. These included the organisation of awareness-raising campaigns and public events to shed light on various topics (e.g. invasive species consumption promotion, importance of the small-scale fisheries sector) as well as the organisation of underwater photo competition, outreach video, outdoor interactive activities bringing the general public on fishing vessels. A series of "learning exchanges" gathering MPAs managers, small-scale fishers and scientists were also organised throughout the project to share best practices and collect ideas to enhance the effectiveness of the awareness raising mechanisms developed.
Who?	Stakeholders that are mobilised in project activities include (small-scale) fishers, other stakeholders representing different marine/maritime activities, local communities as well as managers of MPA.
How?	The FishMPABlue2 project was co-financed by the European Regional Development Fund as part of INTERREG. The FishMPABlue2 involved eight project partners (Federparchi, MedPAN, CoNISMa, WWF Adria, University of Nice Sophia Antipolis, WWF Mediterranean, IUCN Mediterranean, APAM). Within the FishMPABlue2 project, MPAs practitioners were invited to choose which approach to follow in terms of governance to address the main challenges they faced in their MPAs, together with local stakeholders. Management tools related to these challenges were provided and some of them selected by MPAs managers and local stakeholders, associated within pre-existing committees/working groups or within a dedicated "Local Governance Cluster" (LGC). Each pilot MPA received modest financial support to establish a LGC. Governance tools were implemented in pilot MPAs under the coordination of the MPA managing body and with the support of the relevant project partner. Scientific monitoring was carried out before and after implementing the governance tools in order to assess the potential impacts of the governance tools at the ecological, economic and social levels. In the last phase, a "know-how" exchange activity involving all pilot MPAs was carried out to capitalise and transfer the results of the individual implementation of the governance tools, which were then communicated and disseminated trough communication tools, exchange visits, one training course, one conference, national levels events and one online platform.
What lessons?	Reinforcing the engagement of fishers in MPAs management was beneficial in many respects: • Closer work between fishers and MPA managing bodies helped build trust and a shared vision for MPAs; • MPA management bodies and the decisions taken were better perceived by fishers. The overall positive perception of the tools and rules is essential, as it can promote pro-environmental behaviour; • The creation of a Local Governance Cluster involving all actors ensured the needs of local communities were well understood and taken into account. It led to greater support from fishermen and local communities including in complying with decisions taken, therefore facilitating their enforcement and improving the effectiveness of MPAs; • Fishers' engagement can generate an increased sense of ownership and awareness of MPAs; • Some activities (e.g. engagement in surveillance or monitoring) can generate additional incomes for fishers; • Stakeholders' involvement offers an opportunity for knowledge co-production and for the development of participatory research projects that include local knowledge and perceptions, thereby making sure the real local issues and challenges are addressed;

	The FishMPABlue2 project also helped identify conditions that can facilitate the engagement of fishers in decision-making: • Fishers and other stakeholder groups must be provided with some **capacity building** to increase knowledge on participatory processes and on activities they will be engaged in; • **Equal participation** must be encouraged, ensuring that both men and women as well as often-marginalised groups are able to participate. The project stressed some challenges in further engaging fishers in MPA governance: • Fishers can face difficulties linked to peer-to-peer enforcement; • There can be difficulties reaching an agreement on measures that need to be implemented to protect the resources. ***Lessons for the MSFD implementation…*** Illustrations on how to mobilise fishers can be source of inspiration for the mobilisation of stakeholders in the development and implementation of the PoM of the MSFD, complementing in particular activities aimed at more general stakeholder consultation. • The **engagement of all parties in the MSFD governance**, including the most vulnerable groups, as well as the selection of "good representatives" of each stakeholder group, are crucial elements to ensure an agreed upon and effective implementation of the MSFD; • For participatory processes carried out under the MSFD to be successful and inclusive, stakeholders and in particular more vulnerable groups must be **provided with the appropriate means and conditions to participate**. This can be done through trainings and capacity building, or by making sure the participatory tools used are accessible by anyone; • Fishers, as well as other stakeholders, can contribute to MSFD implementation activities such as monitoring (in particular labour-intensive **monitoring activities**). Beyond engaging them into implementation process, such activities can also provide additional sources of income for small-scale fishers.

Civil society involvement and social acceptability of marine energy projects: the MERIFIC project **Scale:** **Policy cycle step**: Implementation of measures, governance and decision-making processes	
Where? What context?	Marine energy development is, among other reasons, conditioned upon the social acceptability of such projects. Developers must work closely with local communities and stakeholders to make sure local concerns and needs are considered in the design and implementation of their projects. The issue of social acceptability is also of relevance when developing aquaculture projects and projects aimed at enhancing marine conservation (e.g. MPAs). The Marine Spatial Planning Directive (MSPD) requires Member States to consult the relevant stakeholders and authorities, and the public concerned, at an early stage in the development of such projects (Article 9)[152]. The Marine Energy in Far Peripheral and Island Communities (MERIFIC) project[153] (2011-2013) was an EU project linking Cornwall (England) and Finisterre (France) that sought to advance the adoption of marine energy in these two regions. One of the objectives of the project was to share best practices and experiment new methods of stakeholder engagement to secure wider understanding and acceptance of marine energy projects (in particular technologies harnessing wave, tidal and current energy and offshore wind). Best practices were built on European examples (11 sites were reviewed in total in France, Spain, UK and Denmark, both offshore and onshore)[154]. Other more recent projects exist, such as UNITED[155] or ENCORE[156]. The UNITED project, focusing on multi-use platforms, has assessed social impacts and acceptability for different pilot sites.
What?	**Social acceptance** of marine projects by local communities as well as **civil society involvement** have received particular attention in the MERIFIC project. It investigated also issues of **awareness-raising** (demonstrating the value of marine energy projects to stakeholders) and of fair **benefit-sharing between project developers and local communities**. Three main types of strategies were deployed to engage stakeholders in the sites studied:

[152] EUR-Lex - 32014L0089 - EN - EUR-Lex (europa.eu) (Last viewed on 20/03/2023)
[153] MERiFIC | Marine Energy in Far Peripheral and Island Communities (Last viewed on 01/12/2022)
[154] GARZON, A., RABUTEAU, Y. and STANLEY, K., 2013. Civil society involvement and social acceptability of marine energy projects – Best practices of the marine energy sector. A report prepared as part of the MERIFIC project "Marine Energy in Far Peripheral and Island Communities". https://tethys.pnnl.gov/sites/default/files/publications/MERiFIC-6.1.2-Civil-society-involvement.pdf
[155] About (h2020united.eu) (Last viewed on 20/03/2023)
[156] https://www.interreg2seas.eu/nl/ENCORE (Last viewed on 20/03/2023)

	• Through information on ongoing development *(information*); • Through involvement in the decision-making process (*planning participation*); • Through stakeholder engagement with financial involvement in the project (*financial participation*), turning stakeholders into shareholders of the projects developed. How to involve stakeholders and inhabitants into marine energy projects is illustrated by the successful experience from the Samsø island in Denmark that commissioned offshore windfarm in 2003. This offshore windfarm provides electricity and financial reward to the local community through a cooperative. The project resulted from a bottom-up initiative that emerged in 1997, with the island elected body deciding to become self-sufficient in renewable energy and opting to develop wind farms for this purpose. To achieve stakeholder engagement, a community cooperative was created, supported and structured by the industry and the local authority. There are other successful examples in Denmark, which is among the main offshore wind power producers in Europe.
Who?	The main stakeholders that are the focus of this initiative are citizens and stakeholders from territories hosting marine energy projects and developments, as well as elected representatives and technical services of local communities.
How?	The implementing entities were the developers of marine energy sites/projects as well as local authorities. Three types of strategies were (and still are) deployed to involve stakeholders: • Information is shared through public meetings and a website. The liaison is then ensured by a communication officer who provides daily updates online; • Early information is provided and goes beyond legal requirements (e.g. the need to have formal public consultation) in setting out proposals. In addition to the environmental impact assessment required, a social impact assessment is carried out. A capacity building group is created to assist the community in participating in the technical aspects of the planning process; • The community owns the offshore windfarm together with Samsø municipality and private owners. Both compulsory and voluntary benefits go into the local community; energy and profit provision goes into the local community, and local businesses are favoured. This effective way of involving local stakeholders and communities is seen as a key ingredient that contribute to, and ensure, social acceptability.
What lessons?	To enhance social acceptability, it is essential to: • Initiate stakeholder engagement **as early as possible**; • **Make relevant information accessible** to all stakeholders, including information on potential (positive and negative) impacts and benefits; • **Go beyond minimum (and legal) requirements** in terms of consultation, in particular having a **stable and inclusive platform or forum** where all parties can exchange knowledge and views that facilitate an inclusive project; • Exploit **opportunities that can be seized by local communities** in terms of new activities developed, jobs, contribution to the territorial development, etc. Overall, such actions contribute to building a common knowledge base to foster a sense of ownership, buy-in and enhance benefit sharing.

	Some challenges and points of particular interest must be paid attention: • Some stakeholders may be difficult to engage in consultation and participation processes; • Deploying a given consultation and mobilisation strategy should not prevent from implementing other ones. For example, providing information does not exclude, but rather supports, the need to involve stakeholders in decision-making processes; • Strategies must be tailored to the local context and the (local and participation) challenges faced, accounting in particular for possible synergies with parallel consultation and stakeholder mobilisation processes organised by local authorities for addressing connected challenges and issues (e.g. wider territorial development, or the need to protect a marine ecosystem). ***Lessons for the MSFD implementation…*** Enhancing the effectiveness of the MSFD implementation could build on better knowledge and appropriation of marine stakeholders and wider society thanks to **participatory processes that go beyond the minimum legal (information and consultation) obligations**. Community organisations and local authorities could contribute to these processes by organising and facilitating **community working groups**.

Social issues addressed in valuation studies supporting the MSFD implementation: a comparative study in the Baltic Sea **Scale:** Baltic Sea **Policy cycle step**: Diagnosis	
Where? What context?	Under the MSFD, Member States are expected to provide an estimate of the **cost of degradation** of the marine environment, i.e. the foregone economic benefits if Good Environmental Status (GES) is not reached. Member States must also perform a **cost-benefit** analysis of new measures needed to achieve GES. To carry out such analyses, monetary values of benefits obtained by beneficiaries of ecosystem services or wider societal benefits are estimated through so-called **valuation studies.** These can build on contingent valuation or choice experiment citizens' surveys that help calculating a mean estimate of citizens' willingness to pay for improvements in the health of marine ecosystem and of the services they deliver. Monetary valuation surveys build on questionnaires that address a wide range of issues including: citizens' use of marine ecosystems; citizens' understanding of marine challenges and problems; citizens' perceptions of who should act to solve problems and why. While monetary values obtained from such surveys are sometimes reported by Member States (MS) (e.g. as benefits that would result from the implementation of the proposed MSFD Programmes of Measures), less attention is given in reporting to the understanding such surveys provide on the relations between citizens and the sea. What such surveys can provide is illustrated here with surveys from Latvia, Finland and Sweden that have been carried out within the MSFD implementation process.
What?	The citizens' surveys carried out in Latvia, Finland and Sweden help better understanding several social-related issues, in particular **citizens' perceptions** of the state of the sea and of marine-related challenges, their **use of the sea** (e.g. recreational activities) and the **connection between the health of marine ecosystems and their welfare/well-being**.
Who?	The main stakeholders targeted by the surveys are citizens from Latvia, Finland and Sweden.
How?	In Finland[157], a survey targeting Finnish citizens was organised in 2017. Before sending the questionnaire, it went through an intensive review by valuation experts. Interviews among ordinary people were conducted. Three different pilot versions of the questionnaire were then tested among the target population before the final version was finally submitted. The survey was sent electronically to 4,999 respondents in Finland, and 1,007 responses were received in total. Besides monetary-related questions, the survey contained questions on: • The previous Baltic Sea use and subjective perceptions of the status of the sea of respondents; • The respondents' concern about the status of the Baltic Sea; • The challenges related to the Baltic Sea (based on a list submitted to respondents, they had to evaluate their degree of knowledge and familiarity of these challenges); • The relative importance of the GES descriptors and the cultural ecosystem services according to Finnish citizens. In Sweden[158], a survey was organised in 2019. It included questions related to:

[157] NIEMINEN E., AHTIAINEN H., LAGERKVIST C-J., OINONEN S.,2019. The economic benefits of achieving Good Environmental Status in the Finnish marine waters of the Baltic Sea. Marine Policy, 99, 181-189, https://doi.org/10.1016/j.marpol.2018.10.014

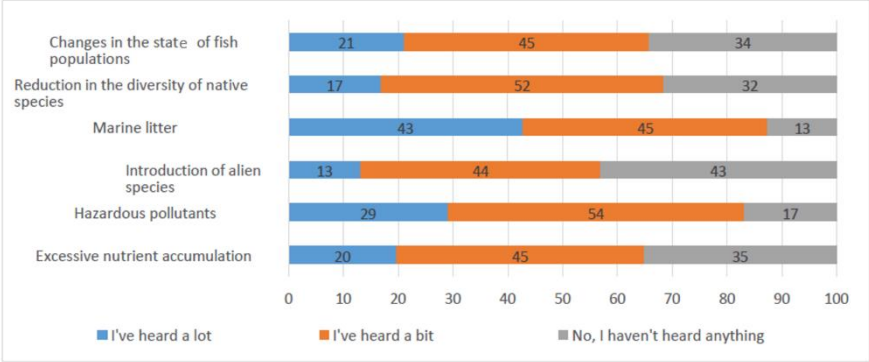
[158] NORDZELL H., WAHTRA J., HASSELSTRÖM L., WALLSTRÖM J. (2020) Värdet av att uppnå god miljöstatus i svenska havsvatten: En betalningsviljestudie. Anthesis. Rapport 2020:8.

	• The respondents' connection to the sea: distance of the living area from the sea, access to holiday home by the sea, degree of concern about the marine environment, last time respondents went to Swedish seas in their free time, marine activities they engaged in in the past year, marine factors that interfered with their stay, perceptions regarding the effect of the sea on health and stress level…; • Attitudes and knowledge about the marine environment: related importance of different areas related to the Swedish coast and marine waters (e.g. recreational activities, habitats for plants and animals, culturally important sites…), attitude towards and knowledge of the sea and its environmental problems. In Latvia[159], a survey targeting Latvian citizens was organised in 2021. The valuation study was implemented as part of a national research project "Improving knowledge about the marine environment" benefiting from EMFF financing. The survey materials, including a questionnaire, were prepared by environmental economists, using, among other, information prepared for the national updated programme of measures (e.g. assessments of the current and business-as-usual-scenario state of the national marine waters, environmental targets and gap analysis). The draft questionnaire was reviewed by marine scientists, professional social and market survey organisers, as well as selected ordinary people. The survey materials were pretested in a pilot survey among the target population. The final questionnaire was applied on a representative sample of the national population, collecting 705 responses by a professional survey company via face-to-face and web-based interviews. The questionnaire included, among other, questions related to: • The respondents' recreational use of the Baltic Sea; • The importance of benefits resulting from interactions with the marine environment; • The respondents' experience of negative impacts and previous knowledge of different marine environmental problems; • Importance of the valued marine environmental problems. Environmental economists are usually involved in the design of such surveys. It is rare, however, that they get support from experts in social sciences (e.g. sociologists or psychologists).
What lessons?	The three studies provided a wider perspective regarding the citizens' perceptions of the importance of marine environmental problems and marine protection. The results showed in particular which issues citizens felt the most concerned about (e.g. litter, hazardous substances, eutrophication issues). Their perceptions seem in line with problems they are the most likely to face, feeling more concerned by issues that are

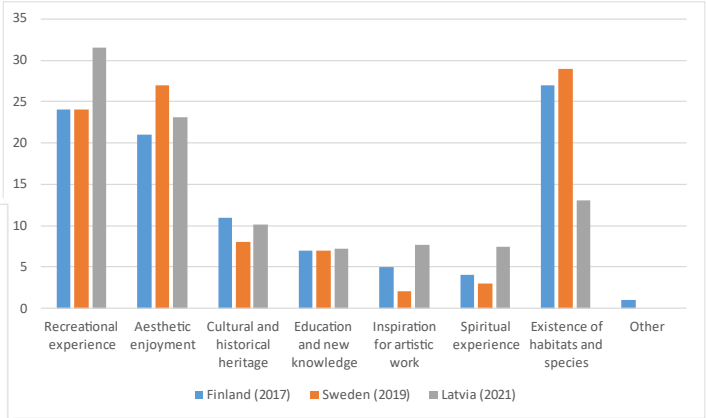
[159] AKTiiVS, 2022. Vides ekonomiskās (monetārās) novērtēšanas oriģināl-pētījums par ieguvumiem no pasākumu ieviešanas laba jūras vides stāvokļa panākšanai. Report to VARAM of a project "Improving knowledge about the marine environment" (Nr. 17-00-F06803-000001) funded by EU EMFF. Available at https://www.varam.gov.lv/lv/media/32243/download

easily noticeable and have a direct effect on sea use and marine activities. Results also showed the most valuable "cultural ecosystem services" provided by the marine ecosystem to citizens.

This type of survey makes it possible to present marine challenges to a large population. Some problems are known by a large part of the population (e.g. marine litter), while others are little known (alien species introduction, changes in fish population).



Example of a valuation study's result: Respondents' previous knowledge of the marine environmental problems in the Latvian marine waters included in the assessment[160]



Example of result of valuation studies from Finland (Nieminen et al., 2019), Sweden (Nordzell et al., 2020) and Latvia (AKTiiVS, 2022): Relative importance of marine cultural ecosystem services (allocating points out of 100 among the individual CES according to their importance)

The fact that the marine environment and its components are relatively unfamiliar to most people as well as complexity of the environmental processes may come as a challenge for survey conception. To cover all relevant environmental problems of the national marine waters, the studies grouped the 11 GES descriptors into six categories (i.e. biodiversity, non-indigenous species, fish stocks, eutrophication, hazardous substances and physical impacts in the Finnish study). Each environmental problem and changes in the state were described using simple wording that can be understood by ordinary people. The change in the environmental state were also presented in a tabular format to facilitate the respondents' assimilation of information. There is a need for such surveys to provide information to respondents about issues unfamiliar for them in a simple and understandable way.

[160] AKTiiVS, 2022. Vides ekonomiskās (monetārās) novērtēšanas oriģinālpētījums par ieguvumiem no pasākumu ieviešanas laba jūras vides stāvokļa panākšanai. Report to VARAM of a project "Improving knowledge about the marine environment" (Nr. 17-00-F06803-000001) funded by EU EMFF. Available at https://www.varam.gov.lv/lv/media/32243/download

	Insights from the inside: results from an interview with a Latvian representative involved in valuation studies Valuation studies' surveys can provide additional information on citizens' perceptions of the state of the sea and of marine-related challenges, their use of the sea and the link between marine ecosystem (e.g. via recreational ecosystem services) and their welfare/well-being. This additional information is provided to policymakers to inform about understanding, views and preferences of the general population. This information is used as a support to policymaking, but usually does not lead to specific measures (e.g. literacy and capacity building measures) in the Programme of Measures. Although valuation studies do not specifically focus on social issues, they can provide relevant information about perception and social impacts of policies from the perspective of the general public. Policies are most commonly designed in collaboration with relevant stakeholders while the views of citizens and of the general public are often missing, mainly as a result of lack of financial resources to carry out national surveys. To ensure data of sufficient quality for policy support, surveys should be based on representative (national) samples (in terms of size and structure) – and this can be costly. Therefore, policy development relies on contributions from stakeholders that are expected to reflect to some level citizens views. Another challenge is that citizens have limited information and knowledge about the marine environment, especially the parts that are inaccessible to them (e.g. benthic habitats). It is important as part of surveys to provide sufficient and well-translated & understandable information on complex environmental matters. For instance, the ecosystem services' approach is a useful tool for explaining the importance of components of marine ecosystems (e.g. benthic habitats) and the need for their protection. With sufficient and understandable information about the marine environment and ecosystem services delivered, valuation studies can also contribute to educating and awareness raising of the general population.
	Lessons for the MSFD implementation… Valuation studies are based on a specially designed surveys, thus provide an opportunity to collect relevant information on public connections to, perceptions of and attitudes toward the marine environment and its protection. Beyond the monetary values they help estimating, citizens' surveys can help better understanding wider society's perceptions about the marine environment and its components, their degree of knowledge on these subjects and ultimately evaluate to what extent citizens (including those that do not use the sea) support marine protection measures. Citizens' survey results can also help designing communication products.

Ocean Literacy in the UK: a multidimensional social survey **Scale:** National **Policy cycle step**: Diagnosis	
Where? What context?	Ocean Literacy, defined as "an understanding of the ocean's influence on you and your influence on the ocean[161]", has gained momentum at different scales - **global** within the framework of the UN Decade for ocean science[162]; **European**, with the launching of the EU4Ocean coalition[163]; national and local scales, with many educational, awareness raising and communication initiatives that aim at enhancing citizens' ocean literacy. In line with one of the objectives of the UK Marine Strategy[164] to enhance ocean literacy, Defra and the Ocean Conservation Trust and Natural Resources conducted a nation-wide study entitled "**Understanding Ocean Literacy and Ocean Climate-related Behaviour Change in the UK"**[165]. This was driven by the belief that enhancing and understanding Ocean Literacy is key to delivering action on climate change, achieving net zero pollution and supporting the UK Marine Strategy. Under the umbrella of this project, a survey on ocean literacy was conducted in 2021 with the aim to understand the current level of ocean literacy in the United Kingdom (UK). This stems from the conviction that ocean literacy is not only about increasing public awareness, but also about providing tools and approaches to transform ocean knowledge into behaviours and actions that promote ocean sustainability.
What?	The UN Decade for Ocean Science mentions several inclusive values relevant to the ocean literacy framework, namely **Inclusion, Diversity, Collaboration and Equity**. It also recognises the role **knowledge, attitude, skills, values, alternatives and barriers** can play in the transformation required to address climate and ecological challenges. Building on this and available experiences presented in the literature[166], the survey explored **Awareness, Knowledge, Attitude, Communication, Behaviour and Activism**, as well as **stewardship, environmental citizenship, marine/ocean citizenship[167], nature and ocean connectedness, access and participation** (in particular exploring improvement of public engagement with environmental issues[168]), as well as **pro-environmental behaviour change** (addressing relationships between (a) internal

[161] https://ioc.unesco.org/our-work/ocean-literacy
[162] UNESCO-IOC (2021) Ocean Literacy within the United Nations Ocean Decade of Ocean Science for Sustainable Development, A framework for Action.
[163] The EU4Ocean coalition is supported by DG MARE and continues with its 2nd phase for another three years to promote ocean literacy across Europe and its sea basins. https://webgate.ec.europa.eu/maritimeforum/en/node/4484
[164] https://moat.cefas.co.uk/introduction-to-uk-marine-strategy/
[165] This study was conducted by Defra - UK Department for Environment Food and Rural Affairs - in collaboration with the Ocean Conservation Trust and Natural Resources Wales.
[166] Brennan, C., Ashley, M. & Molloy, O., 2019. A System Dynamics Approach to Increasing Ocean Literacy. Front. Mar. Sci. 6:360. doi: 10.3389/fmars.2019.00360
[167] "***Marine citizenship*** *can be defined as having understanding of the individual rights and responsibilities towards the marine environment, having an awareness and concern for the marine environment and the impacts of individual and collective behaviour, and having a desire to have a role in ensuring on-going sustainable management of the marine environment* " in (McKinley, E., 2010. A Critical evaluation of the application of marine citizenship in sustainable marine management in the UK. Doctoral Thesis (Doctoral). Bournemouth University.)
[168] Arnstein, S.R., 1969. A Ladder of Citizen Participation. Journal of the American Planning Association, 35(4), pp. 216-224.

	factors – emotions, social values and (b) external factors – socio economic context, political drivers … on individual behaviour change[169]). It was proposed also that the survey investigates: **emotions** (emotional connection, empathy, fear…) considered as having a central role in driving behaviour change; **access** to information but also to the ocean itself; **social values**[170] (both monetary and non-monetary) linked to natural capital; **motivations** (understood as the reasons "why" people would be willing to make certain decisions and act in a specific way). Such components have helped understand if people are willing to act following a top-down initiative, the application of fines or any other mechanisms aimed at influencing their practice and behaviour. **Trust and transparency** were also considered in relation to the sources of information people use and their reliability, as these are factors influencing individuals' desire to learn more about ocean related topics thereby shaping public attitudes towards marine issues. As a result, building on supporting evidence from the literature, the final survey investigated 8 dimensions: **Awareness, Knowledge, Attitude, Communication, Behaviour, Activism, Personal/ emotional connection and Access, experience, and proximity.**
Who?	The survey targets citizens from England, Wales and Scotland, aiming at capturing the views from citizens representative of the population diversity of these countries.
How?	The survey built on **online panel method** to reach a maximum of respondents through multiple panels, with the ability to complete the survey on tablet and mobile devices. The collection of results followed a series of fieldwork activities, and the age/region quotas were set to achieve representativeness of the population, based on the Office for National Statistics Coastal Communities. In total **12 166 interviews** were achieved across England, Wales and Scotland. The survey consisted of **5 sections,** with a mean duration of around 22 minutes per questionnaire response. Results were presented in categories addressing different dimensions of ocean literacy, some results being illustrated below.

[169] Stoll-Kleemann, S., 2019. Feasible options for behavior change toward more effective ocean literacy: a systematic review. Front. Mar. Sci., 6:273.
[170] An interesting example worth mentioning is the Marine CoLAB Community of practice putting together marine NGOs and centered about "values" of the ocean. (https://marinecolaboration.wordpress.com/)

Awareness	
% believe health of marine environment is very good/good	Global 12% England and Wales 17%
% with very good/good awareness of global challenges	32%
Knowledge	
Marine terms with some understanding	Climate change 93% Sustainable fishing 85% Sustainable development 73%
Attitudes	
Top 3 benefits from marine environment	Plant and animal habitats 53% Food 40% Weather & climate control 31%
3 highest threats to the marine environment	Litter and plastic pollution 74% Chemical pollution 57% Overfishing 54%
% protecting marine environment very important/ important	85%
% Human activity contributes to climate change	49%
Activism	
Top 3 pro-marine acts	Lifestyle changes 39% Petitions 26% Membership of organisation 15%
Behaviour	
Top 3 pro-climate behaviours	Recycling 81% Reduced plastic 73% Reuse plastic 68%
Have or plan on making lifestyle changes	78%
Reasons for changes	Concern about climate 69% Desire to be greener 68% Care for marine environment 63%
% believe lifestyle has impact	53%
Communication	
Sources of knowledge about the marine environment	Television/radio 48% Films/Nature/Wildlife 47% News 46%
Emotional / personal connection	
Top 3 emotional terms	Concern 49% Awe/wonder 42% Curiosity 28%
Access, experience & proximity	
Never visited the marine environment	11%
Top activities	Walking 62% Photography 31% Dog walking 27%
Outcomes of visits	Mental health 84% Physical health 80% Time with others 56%
Ocean Literacy Principles	
% Mostly or completely true	The ocean supports a great diversity of life and ecosystems 92%
	The ocean is largely unexplored 77%
	The ocean is a major influence on weather and climate 83%

Summary of key findings of the UK Ocean Literacy Survey (2021)

When asked what **emotional responses** respondents had to the marine environment, **concern** was most reported feeling (49%) followed by awe/wonder (42%) and curiosity. *This indicates a potential interest by the general public in the quality of the marine environment, and perhaps an eventual implication in environmental actions proposed by policy makers.*

When asked about the **most common sources of knowledge and information** that respondents use to learn about the marine environment answers were TV and Radio (48%), films and documentaries (47%) and news media (46%) while social media was only 23%, and talks/seminars just 3%. *This information is very important to help direct communication strategies, as well as channelling funds and resources towards production of content and information generation.*

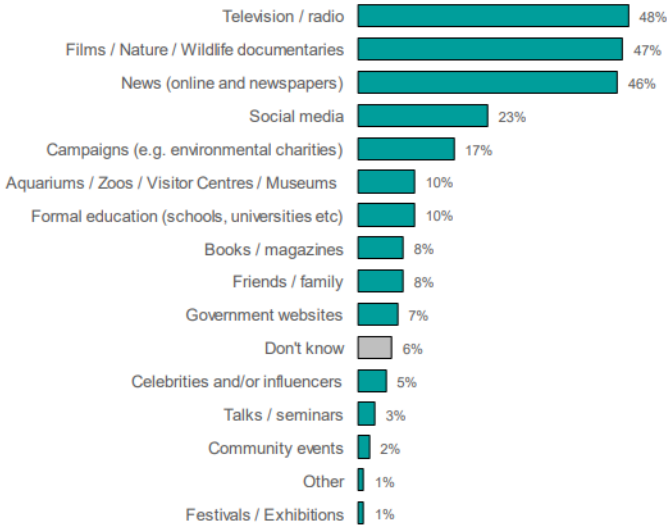
16. Communicating about the marine environment

The most common sources of knowledge/information about the marine environment in the last 12 months was TV and Radio (48%), films, nature and wildlife documentaries (47%) and news media (46%) (Figure 21).

While social media was indicated by 23% of respondents, only 5% indicated a role for celebrities or influencers.

The least common sources were talks/seminars (3%), community events (2%) and festivals/exhibitions (1%).

Figure 21: Sources of knowledge about the marine environment (weighted %)



Q14: Thinking about the last 12 months, where do you think your knowledge/information about the marine environment in England/Wales has mostly come from? Please select a maximum of three
Unweighted base: 8,440

When asked about **outcomes and motivations of visiting coastal/marine spaces**, mental and physical health were the most frequently reported (84% and 80%). *This information creates a direct link between the respondents and the quality/accessibility to coastal and marine spaces. As such, respondents would be more implicated in policies affecting these spaces as their own wellbeing derives from the quality of these spaces.*

What lessons?	The study[171] and the survey's results[172] allowed to identify **enabling factors that are influencing behaviour change** in relation to ocean literacy in the UK (for example the level of awareness and interest in ocean issues) and **barriers to knowledge,** awareness and literacy (for example using the right sources of knowledge/information to reach the general public). The survey helped **better understanding respondents' willingness to change their behaviours/attitudes** towards ocean and climate change issues (in particular understanding their motivations, their existing habits, and the right ways to guide them towards sustainable actions). It also helped understanding **topics of concern for the general public** (for example sustainable fishing) as well as identifying topics the general public is less or not aware of (for example eutrophication, good ecological status) that can help better targeting public engagement actions and measures. Finally, the survey provided a first **nationwide deep understanding of the population's level of knowledge** of marine issues in general, and of their **perception** of ocean health challenges. ***Lessons for MSFD implementation…*** By better capturing current knowledge, communication channels, perceptions and willingness to adapt, such a citizens' survey can contribute to more **targeted communication** as well as to designing actions that can support **mobilisation and behaviour change and that can be integrated into the Programme of Measures.** It underlines the importance of better understanding how information is produced, shared and accessed. Carried out at different time intervals (e.g. every 6 years, in coherence with the planning cycle of the MSFD), it can also help capturing changes in citizens' perception that result from actions and improvements of the health of marine ecosystems, as well as wider societal changes.

[171] McKinley E. and Burdon D. (2020) UNDERSTANDING OCEAN LITERACY AND OCEAN CLIMATE-RELATED BEHAVIOUR CHANGE IN THE UK - WORK PACKAGE 1: EVIDENCE SYNTHESIS. Final report produced for Ocean Conservation Trust & Defra

[172] Defra, 2021. Ocean Literacy in England & Wales: Headline Findings Report. Defra project ME5239.

Women in the sustainable blue economy: contributing to gender equality in maritime sectors **Scale:** Multiple **Policy cycle step**: Implementation of measures	
Where? What context?	Traditionally, men have dominated marine & maritime sectors and industry, while women represented only a very small percentage of the workforce in the maritime community[173,174]. Working conditions such as **lower wages, gender disparity and discrimination/harassment on board** do not encourage, for example, women to opt for a career at sea[175]. Moreover, the **perception and stereotypes** towards women's participation in maritime activities have not been very favourable, often linked to wider **social and cultural norms**[176]. The lower women's enrolment in Science, Technology, Engineering and Mathematics (STEM) fields[177] contributed also to **lack in capacity building and skills** required for some certain maritime sectors. However, recent years have witnessed positive changes with experiences illustrating the role women can have in maritime professions related to the different building blocks of marine value chains (including for emerging sectors such as blue bioeconomy), marine policy, science, education or ocean literacy. National and international public and private organisations are making efforts to close the gender gap, while the enrolment of women in maritime sectors has gained momentum as illustrated by initiatives at: the **international level** (such as the IMO's Women in Maritime initiative[178], a gender and capacity building programme which puts in place an institutional framework to incorporate a gender dimension in IMO's policies and procedures); the **EU level** (for example EMFAF's call for proposals on Women in the Blue Economy[179] which aims to develop initiatives that can help overcoming constraints to women participation in sustainable blue economy); or **regional (sea basin) and national levels** as illustrated by initiatives supporting women's business development and active participation in the blue economy (e.g. the Union for the Mediterranean's support to activities targeting women in Blue Economy in the Mediterranean[180,181]). In addition to the roles of international organisations and national governments, civil society organisations are contributing to the achievement of gender equality in maritime sectors.
What?	Looking at the gender issues in the maritime sectors **through** the lens of the **UN SDGs 2030 agenda**, the targeted Sustainable Development Goals would be SDG 5 (Gender equality) and SDG 14 (Life below water). From a European perspective, the **2012 Treaty on the EU**[182] envisions a society *"in which pluralism,*

[173] Women represent only 1.28% of the global seafarer workforce (BIMCO/ICS (2021) Seafarer Workforce Report)
[174] Women represent only 24% of the EU Aquaculture workforce (STECF 20-12 (2021) EU Aquaculture sector Economic report 2020)
[175] Arulnayagam (2020) Addressing the Voice of Women in Marine and Maritime Industry http://dx.doi.org/10.31364/SCIRJ/v8.i4.2020.P0420763
[176] A study from a Greek perspective sheds light on this very topic, whereby women's expected role in the family rendered it difficult to pursue an employment on board or at sea, and a majority of seafarers were not willing to accept women as their superior. (Theotokas and Tsalichi (2013) Employment of women at sea. Perceptions, attitudes and experiences of male seafarers in the Greek context.)
[177] Women represent only 24% of STEM graduates (https://www.womeninoceanscience.com/welcome01)
[178] https://www.imo.org/en/OurWork/TechnicalCooperation/Pages/WomenInMaritime.aspx
[179] https://cinea.ec.europa.eu/news-events/news/women-blue-economy-call-proposals-now-open-2022-05-17_en
[180] https://south.euneighbours.eu/news/ufm-emuni-webinar-women-blue-economy-mediterranean/
[181] https://medblueconomyplatform.org/vkc/news/webinar-women-and-blue-skills-careers-jobs-in-lebanon-6d5098ee40/
[182] Articles 2 and 3. https://eur-lex.europa.eu/resource.html?uri=cellar:2bf140bf-a3f8-4ab2-b506-fd71826e6da6.0023.02/DOC_1&format=PDF

	non-discrimination, tolerance, justice, solidarity and equality between women and men prevail" and where the EU itself has committed to *"eliminate inequalities, and promote equality, between men and women".* The EU's commitment to achieving a Union of Equality is reflected in the **Gender Equality Strategy (2020-2025)**[183], and in a more recently adopted **Directive on gender balance in corporate boards (2022)**[184] aiming to improve the gender balance in corporate decision-making positions in the EU. As such, the social issues surrounding the role of women in maritime sectors are plentiful and multi-layered. Starting with **ensuring equal access** to education and training opportunities, to helping women **face barriers** towards achieving a career in the maritime realm, and finally through providing the adequate **working rights and conditions** within the actual job.
Who?	The primary group are women interested in being engaged for the sea or that are already engaged, in particular from younger generations, sea users and the general public. The inclusion of men in the debates and initiatives is also crucial in the role they can play as "male allies" contributing to awareness raising, speaking out against oppression/ sexual harassment and more generally ensuring safe, supportive and attractive working conditions are offered to women to support their career development.
How?	From the large list of initiatives taking place in Europe, three initiatives carried out by civil society organisations fringing international, regional, and local perspectives are presented here – as illustration and source of inspiration. ***Example 1 – WISTA (Women's International Shipping & Trading Association)*** WISTA is an international network of women in maritime sectors[185], and its mission is to attract and support women, at the management level, in the maritime, trading and logistics sectors. The governance consists of an executive committee, elected during the annual general meeting of the association and three thematic committees[186] working together to advance the WISTA mission. To maximize the outreach and the results, WISTA International leads a web of 54 national WISTA associations, with appointed presidents for each of the local associations. The adherence to these local associations is membership based, with a fee that contributes to the various activities' organisation, but that also boosts commitment, engagement, and the sense of belonging of its members. The focus of the organisation is **professional development** of women in maritime sectors, and therefore targets the building of a global network of **business relationships** and supports the advancement of women in **leadership roles** to close the gender gap. This is achieved through **capacity building** activities, **business and skill building** opportunities, as well as **corporate visibility** and facilitation of relationships between members and the maritime industry.

[183] A Union of Equality: Gender Equality Strategy 2020-2025. (https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:52020DC0152&from=EN)
[184] https://ec.europa.eu/commission/presscorner/detail/en/statement_22_7074
[185] https://wistainternational.com/
[186] The three committees are Diversity, Human Resources and Future and Technology committee. The latter has its own blog posts, accessible on https://wistainternational.com/tfc/

	In 2020, WISTA has signed a **MoU** with the International Maritime Organisation (IMO) agreeing to **collaborate** and develop areas of cooperation in line with both organisations' axes of priority[187]. In collaboration with the IMO, WISTA also contributes to **building knowledge on gender diversity in the maritime sector,** such as the 2021 **"Women in maritime survey report"**[188] that contains information on the proportion and distribution of women working in the maritime sector of the IMO's member states[189]. WISTA also contributed to the publication of "**Gender Diversity Handbook**"[190] which is based on an online survey designed to examine female seafarers' perceptions and manifestations of discrimination on board ships[191]. This report also addresses policies implemented by the maritime companies to help fight such gender-based harassment and bullying[192]. Another fruit of the collaboration with IMO, as well as Maritime SheEO, is the establishment of a "**Leadership Accelerator Programme"** (LEAP)[193] with the following objectives: offering management knowledge and skills to help women in maritime industry move to leadership roles and enable women to become part of a network of female executives to support and inspire each other along the leadership journey. ***Example 2 – WIMA (IMO) (Women in Maritime Associations)***[194] Born under the hospices of the IMO through its gender and capacity building programme, eight Women in Maritime Associations (WIMAs) have been launched[195] in Africa, Arab States, Asia, the Caribbean, Latin America and the Pacific. These associations are platforms to discuss issues beyond gender, including **technical aspects relevant to the industry**. The activities endorsed are the organisation of maritime career days, beach clean ups, awareness raising etc. Other results of these associations' establishment are **publications and knowledge generation**[196], development of **regional strategies for**

[187] WISTA also has a consultative status with the IMO. (https://wistainternational.com/news/wista-international-signs-mou-with-the-international-maritime-organisation-imo/)

[188] http://wistainternational.com/wp-content/uploads/Women-in-maritime_survey-report_high-res.pdf

[189] This report reveals that women account for only 29% of the workforce in the maritime industry; only 2% of seafarers are women and are mostly in the cruise industry. (https://wistainternational.com/our-work/women-in-maritime-imo-wista-international-survey-2021/)

[190] http://wistainternational.com/wp-content/uploads/Gender-Diversity_bookletSize-reduced.pdf

[191] This handbook highlights that 60% of women reported gender based discrimination onboard, 66% reported harassment and intimidation by male employees, and 25% reported that physical and sexual harassment is common onboard. (https://wistainternational.com/the-diversity-handbook/)

[192] For example, 97% of respondents declared that their companies had harassment and bullying policy.

[193] https://wistainternational.com/resources/

[194] https://www.imo.org/en/OurWork/TechnicalCooperation/Pages/WomenInMaritime.aspx

[195] in Africa, Arab States, Asia, the Caribbean, Latin America and the Pacific.

[196] For example, a list of publications developed by the Network of Women of Maritime Authorities of Latin America (Red-MAMLa) (https://www.redmamla.org/es/cronicas-maritimas)

women in maritime sectors as well as development of monitoring and evaluation schemes [197], establishment of regional networks supported by social media.

To create and contribute to a sustainable maritime community by providing a more inclusive, safe working environment, equal opportunities, and fair treatment for all Pacific women in the maritime sector							
Long-term Outcomes	**Key Results**	**Indicators**	**Activity 1**	**Activity 2**	**Activity 3**	**Activity 4**	**Comments**
Improved leadership, recognition and contribution of women in the Pacific maritime sector	**Result 1.1: Maritime community employers, stakeholders and government entities recognize and implement the provisions contained in international treaties pertaining to the equal treatment, empowerment and protection of women in the maritime community**	**Indicator 1.1.1:** Number of PICTs, maritime sector employers and other stakeholders with evidence of adoption and enforcement of relevant treaties to promote equal treatment, empowerment and protection of women in the maritime sector	Conduct a stock-take and review of PICTs' national frameworks and policy for gender equality	Establish contact with maritime administrations and relevant stakeholders in PICTs to outline their respective responsibilities and potential influence over the development of policies and systems relating to all aspects of gender equality with a MoU for the formal recognition of Women in Maritime Associations	Participation of PICTs' policy development institutions and maritime administration directors (irrespective of gender) in regional conferences for PacWIMA	PICTs' maritime administrations to nominate women in maritime to attend training in WMU, IMLI, Lloyds Maritime College, University of Tasmania and recognized maritime training institutes	Important to know channel of communications in PICTs to address issue for enforcement and follow-up
		Indicator 1.1.2: Number of professional women employed in the maritime sector	Analyze base-line data and desegregate, collected through PacWIMA hub to identify age, wages and salaries and level of professional development	Identify and mobilize champions (male/ female) to raise awareness, support and report on the role of women in all capacities	Network and promote employment opportunities for women at professional level in government and the private sector		

Section of the results framework developed within the Pacific Women Strategy to monitor performance

These associations' initiatives play a role in **narrowing some of the institutional barriers**, and most importantly the **cultural stigma** facing women who enter the maritime industry.

Example 3 – Women for sea association[198]

Established in 2010, the main topics that the Women for Sea association addresses are **the integration of women in maritime sectors and maritime conservation activities.** Their activities are coupled with **environmental awareness** initiatives aiming for the preservation of marine and coastal ecosystems. Their activities also utilise traditional knowledge to identify sustainable solutions for marine environmental challenges. Its three main axes of action are "**feel**" (to incite emotions and (re)create links with the sea), "**learn**" (to transfer knowledge and marine scientists' work for the benefit of marine and coastal ecosystems by sensitising the public to the impacts of human activities on the sea) and "**act**" (to develop actions for the sea and contribute to the eco-responsible research and management in marine conservation). The outcomes of the various activities reached over 20,000 persons through their

[197] For example, the Pacific Women regional strategy developed by the Pacific Women in Maritime Association (PacWIMA) (https://prdrse4all.spc.int/sites/default/files/t8_-_annex_a_regional_strategy_for_pwm_2020-2024_0.pdf)
[198] https://womenforsea.fr/

	campaigns, and over 200 students, in addition to participating in international conferences (such as One Ocean Summit – Brest). Initiatives carried out by the association include: • **The water drop odyssey:** This project occurs in line with International Oceans Day (8 June) and follows the journey of a water drop from its mountain source to the sea. This project includes **pedagogical activities** (organisation of workshops, sensibilisation campaigns, setting eco-booths within nature reserves...) as well as **sports activities** (organising mountain biking/kayaking along the river basin). Collaborations with schools and teachers is an important part of this project and allowed to reach a large number of educational and local institutions. • **Sustainable boating/sailing awareness:** Through creation of videos on eco-navigation, inclusion of industries/boating sectors in awareness activities, dissemination through mailing lists and publication of monthly articles in the "*Voiles et voiliers*" magazine (This magazine has 25,000 subscribers and thus dissemination efforts are amplified). • **Women for sea odyssey:** This initiative aims to explore the role of islands as models of sustainable development and thus develop an "archipelago" of innovative solutions based on island residents' knowledge. The sailing expedition builds on an **anthropological study** of the perception of islanders of the changing marine environment, on putting forward successful initiatives of women in islands through multimedia portraits, and organisation of educational workshops for the youth focusing on Mediterranean voices of the future. The strong point of this initiative is its **multidisciplinary** dimension: women from different backgrounds such as anthropology, science and marine biology, art… gather to explore means towards a sustainable marine environment. Positive **outcomes** emanating from these sailing expeditions are **collaborations** between women from different sectors (one example is the anthropologist collaborating with the marine biologist, that met on board), but also the **empowerment** women felt through their sailing experiences (members noted that women who participated felt more confident, autonomous and were willing to take more initiatives, which was enabled by the safe environment on board).
What lessons?	The examples presented above stress the need to: • **Establish strong partnerships** with international organisations, national/local authorities, research institutions, as well as private sector and industries (boating companies, tourism operators…), media companies, and more; • **Organise inclusive and interactive activities**, by facing and meeting the challenges alongside mobilised and committed women beyond merely presenting them; • **Build on traditional and local knowledge for identifying sustainable solutions**: bringing in women who have successful stories of sustainable management practice within their communities so these are used as basis for developing new actions and initiatives. ***Lessons for the MSFD implementation…*** • Ensuring a **gender-balanced governance,** mobilisation and consultation of the MSFD implementation process is rarely an objective of public services in charge of the implementation of the MSFD. Still, it can contribute to changing the current gender-imbalanced situation and bring a gender agenda to marine policy and maritime sector development. At the more local level, the involvement of women in processes that help strengthening the traditional and inherited knowledge brought into discussions, as well as bringing experiences on activities predominantly carried out by women; • Bringing a gender dimension to marine policy development and implementation can benefit from **collaborations** with organisations active from the local to international scales that are addressing gender issues and already developing projects and initiatives; • With a longer-term perspective, actions can be taken to enhance gender equality throughout the various phases of marine-related and maritime **careers** (education , job search, employment…).

Assessment of the social impacts of environmental measures taken under the MSFD **Scale**: **Policy cycle step**: Selection of measures	
Where? What context?	In France, the Marine Strategy Framework Directive (MSFD) and the Maritime Spatial Planning Directive (MSPD) are implemented simultaneously through so-called *Documents Stratégiques de Façade* (Façade Strategic Documents or FSD). These documents at the scale of maritime facades include a section that defines strategic objectives for marine waters and resources, as well as an operational section that set up the monitoring programme and the Programme of Measures (PoM) proposed for achieving strategic objectives. Up to 67 new environmental measures have been proposed for the different FSD, with differences between facades depending on their ecological context and marine/maritime activities. An ***ex-ante* assessment of the economic and social impacts** of the measures proposed for each FSD was carried out in 2021. The assessment of the economic and social impacts on marine and coastal activities aimed in particular at assessing the acceptability of proposed measures for different socio-economic activities and thus to evaluate the likelihood of their effective implementation.
What?	Social issues investigated are the **social (and economic) impacts of environmental measures on marine activities**, such as the ease of access to marine resources and the cost (in terms of time and money) of environmental measures for some economic sectors in particular. **Socio-economic measures that could mitigate the negative impacts** of the new environmental measures were also studied. These measures include e.g. the development of a training offer related to marine renewable energy and the development of a dredging sediment re-use channel aimed at supporting operators to limit the impact of dredging on the marine environment and species. Attention was in particular given to: • Assessing the **level of knowledge of professionals** on future environmental measures and how they are getting prepared for them (if at all). Some measures aim in particular to raise sea users' awareness on marine issues (e.g. marine non indigenous species) and to ensure knowledge sharing, making sure the environmental measures are properly justified and that their interest is well understood. For instance, the project of FSD foresees two socio-economic measures to facilitate the sharing and dissemination of all data related to the marine environment. These measures should help to improve recreational fishermen's knowledge of the issues related to the protection of the environment and species, and thus facilitate understanding of the measures to be implemented and their acceptance; • Ensuring the **social acceptability** of measures to some extent (e.g. measure D06-OE02-AN1 "Improving the management of Posidonia beds on beaches by reconciling social acceptability with the issues of erosion management and the protected species"); • Mobilise sea users and professionals as **actors in the protection of the marine environment** by identifying based on consultation more sustainable practices that professionals could carried out accounting for their priorities and constraints. This has also contributed to the build-up of a **collective culture**, in particular through co-construction of the PoM with all sea users.
Who? (stakeholders affected)	The main stakeholders that were associated to the impact assessment are the socio-professional stakeholders of marine and coastal activities, in particular: beach activities and beach use; Aquaculture; Submarine cables; Shipbuilding; Extraction of marine materials; Boating and water sports; Recreational fishing; Professional fishing; Electricity production; Coastal tourism; Maritime transport and ports; and Maritime public works.
How?	The assessment, carried out by consultancies, mobilised national experts in the different marine/maritime activities, representatives of Interregional Maritime Directorates as well as professionals representing the different activities and professions. The new measures proposed under the FSD projects were classified into different categories to assess the relative importance of each measure's social impacts on marine

	activities and to select a list of measures to be analysed in more details. Measures have been classified based on the following typology of impacts: 1) Binding impacts: measures that explicitly envisage restrictions or limitations of activities, or involve investments by private operators; 2) Possible but uncertain and time-lagged impacts: measures that require improved knowledge before they can be precisely defined and whose socio-economic impact is not clear and therefore difficult to estimate at present; 3) Impacts by encouraging changes in practices in the long term: measures that aim to disseminate knowledge among practitioners of the activities by intervening, e.g., in the training of professionals, user awareness, environmental education; or that aim to better organise governance and State action, particularly in terms of control; 4) Long-term impacts on the context of intervention of activities: measures that do not directly target an economic activity but rather public investments that will probably benefit, in the long term, certain activities by improving the resource or the environment. The analysis was then conducted for category 1 and 2 measures considered as having the largest impact in the short term. The analysis of the measures allowed to select a list of activities that would be most impacted for which more detailed impact analysis was carried out. Interviews with experts in these activities were carried out to assess (qualitatively) the order of magnitude of impacts. A first series of worksheets covering seven activities was drafted to take into account of local contexts (types of activities, particular environments…), and their content discussed and consolidated through additional interviews (note: workshops originally proposed could not be organised due to the Covid-19 pandemic).
What lessons?	Several challenges in identifying and assessing socio-economic impacts of the environmental measures were identified: • The difficulty of identifying the effects of proposed measures due to the multitude of external factors (e.g. Brexit, Covid-19 health crisis); • Measures can have cumulative effects on certain activities, therefore the analysis of social impacts of a particular measures should not ignore the existence of other measures. It is therefore more relevant to conduct assessments of social impacts at the economic activity scale accounting for the several measures affecting this economic activity; • The limited precision on the dimensioning of some measures and on the conditions for their implementation made the estimation of social impacts challenging. ***Lessons for the MSFD implementation…*** • Social impacts of proposed measures can be assessed, in addition to environmental and economic impacts, within the scope of multicriteria analyses that can complement cost-effectiveness and/or cost-benefit analyses. This would help selecting measures accounting for potential social impacts as well as accompanying measures for limiting some negative social impacts. Combined with a solid stakeholder mobilisation, this can positively enhance acceptability of some of the measures proposed; • The consultation of maritime professionals in the choice of measures conditions their acceptability and effective implementation. Professionals must be sufficiently informed and consulted. Sufficient attention should be given to assessing short term and long-term impacts, as some measures having negative short-term impacts can eventually delver benefits in the long term – including social and wider societal benefits.